Development of a Diagonal Isometric Tensor Network Algorithm

Master‘s thesis

Benjamin Sappler

Simulation of QMB systems

- Simulation of QMB systems is a hard problem

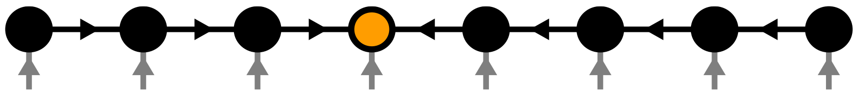
$$|\Psi\rangle = \sum_{i_1, i_2, \ldots, i_N = 1}^{d} \Psi_{i_1, i_2, \ldots, i_N} |i_1\rangle \otimes |i_2\rangle \otimes \cdots \otimes |i_N\rangle$$

- Fully describe wavefunction: d^N parameters!
- Approximations are needed!

Contents

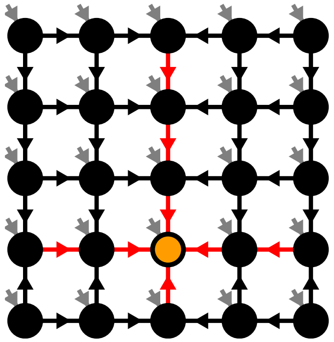
1. Matrix Product States (MPS)

   [M. Fannes et al. 92]

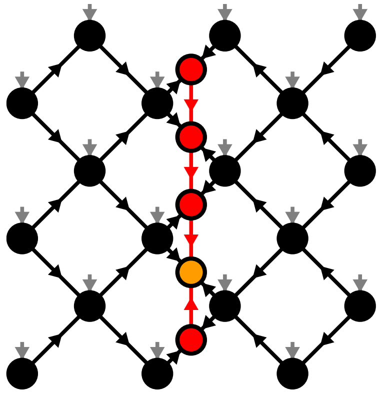


2. Isometric Tensor Product States (isoTPS)

   [M.P. Zalatel, F. Pollmann, 2020]



3. Alternative canonical form



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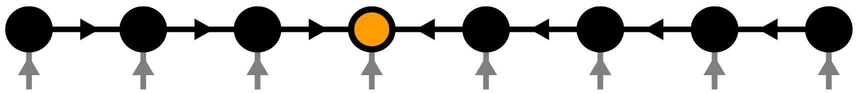


2. Isometric Tensor Product States (isoTPS)

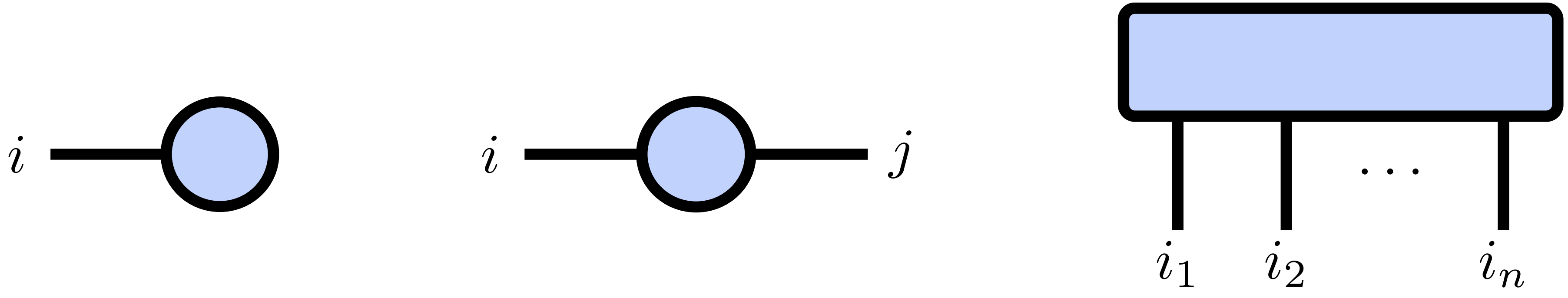
   [M.P. Zalatel, F. Pollmann, 2020]

3. Alternative canonical form

Tensor Networks

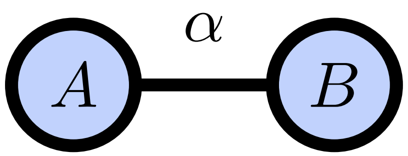
- Tensor of rank n: $T \in \mathbb{C}^{\chi_1 \times \chi_2 \times \cdots \times \chi_n}$

  $T_{i_1, i_2, \ldots, i_n} \in \mathbb{C}$

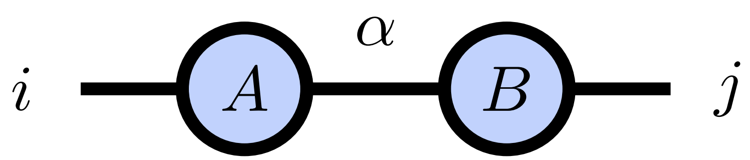


Index Contractions

- Scalar product: $c = \sum_{\alpha=1}^{\chi} A_\alpha B_\alpha$

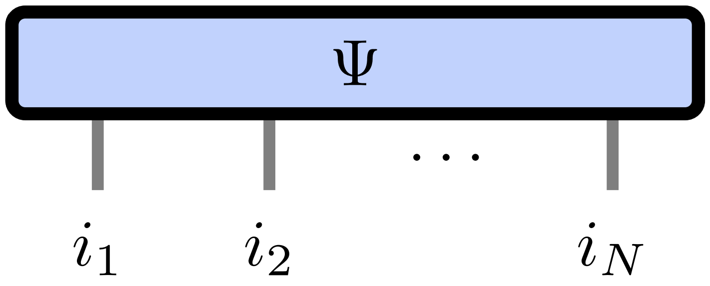


- Matrix product: $C_{ij} = \sum_{\alpha=1}^{\chi} A_{i\alpha} B_{\alpha j}$



Matrix Product States

$$|\Psi\rangle = \sum_{i_1, i_2, \ldots, i_N = 1}^{d} \Psi_{i_1, i_2, \ldots, i_N} |i_1\rangle \otimes |i_2\rangle \otimes \cdots \otimes |i_N\rangle$$



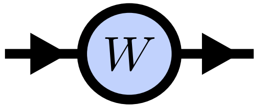
- Number of parameters: $d^N \to N\chi^2 d$
- Represents area law states

χ

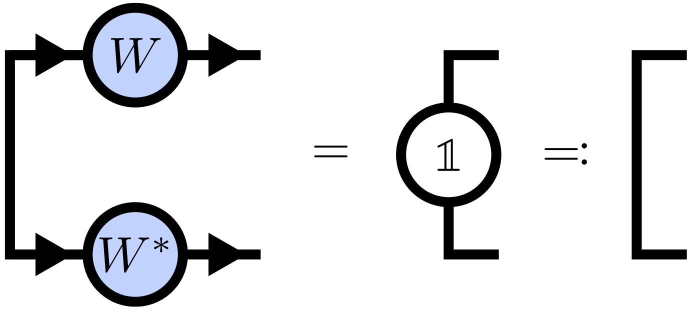
[M. Fannes et al. 92]

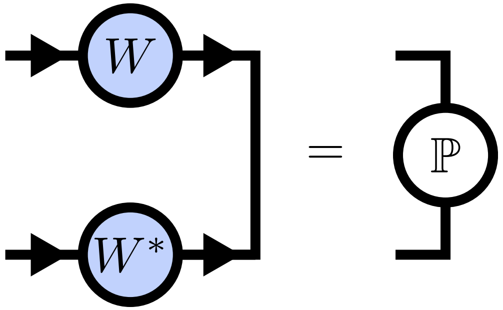
Isometries

- Isometry: $W \in \mathbb{C}^{n\times m},\ n \geq m$

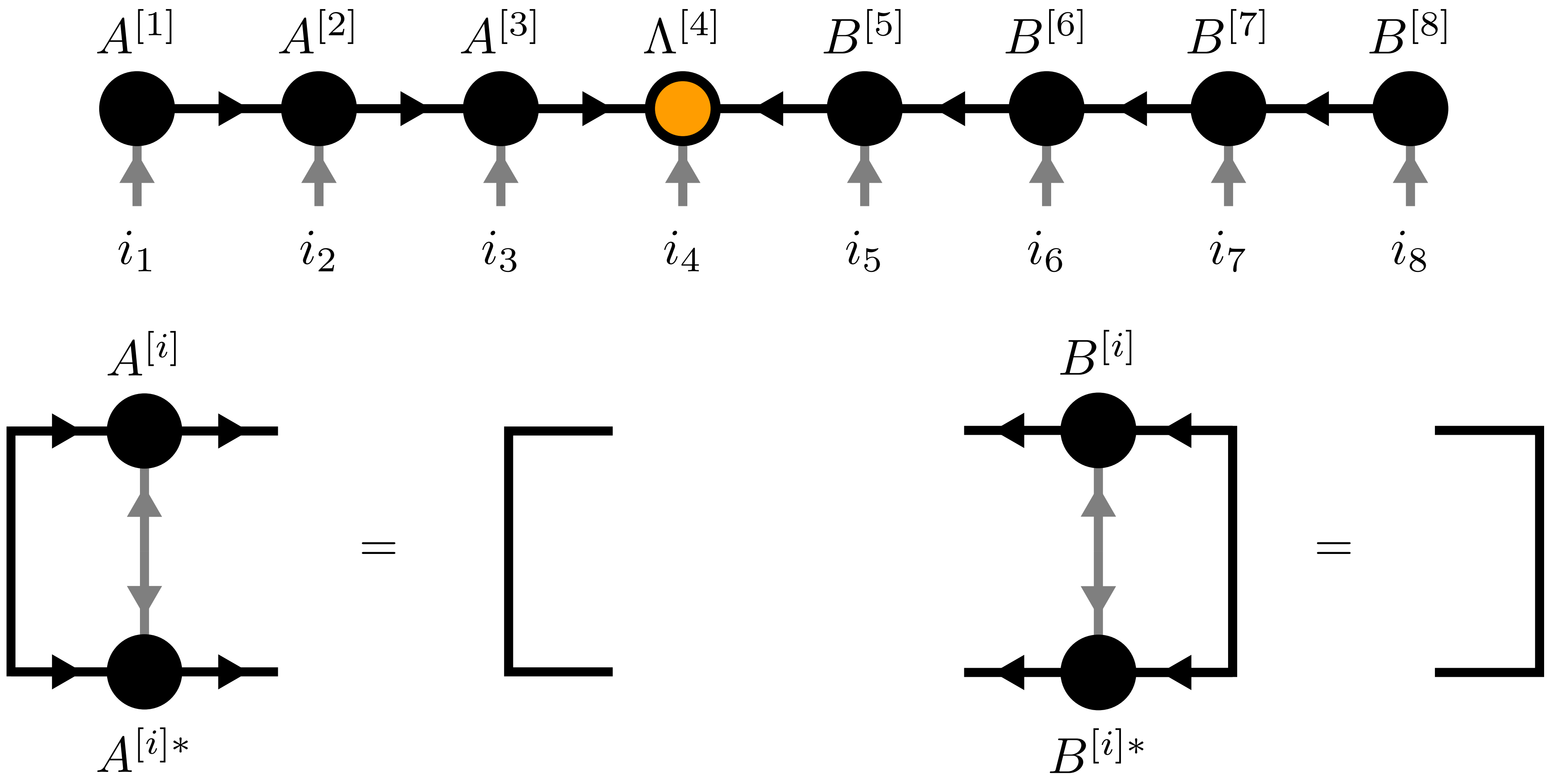


- Isometry condition:
  - $W^\dagger W = \mathbb{1}$
  - $WW^\dagger = \mathbb{P}, \quad \mathbb{P}^2 = \mathbb{P}$



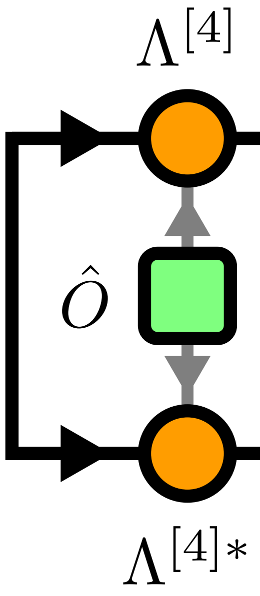


Canonical Form



Local Expectation Values

$$\langle\Psi|\hat{O}_n|\Psi\rangle =$$



Time Evolving Block Decimation (TEBD)

- Schrödinger equation: $$|\Psi(\Delta t)\rangle = \hat{U}\left(\Delta t\right)|\Psi\rangle = e^{-i\Delta t\hat{H}}\,|\Psi\rangle$$

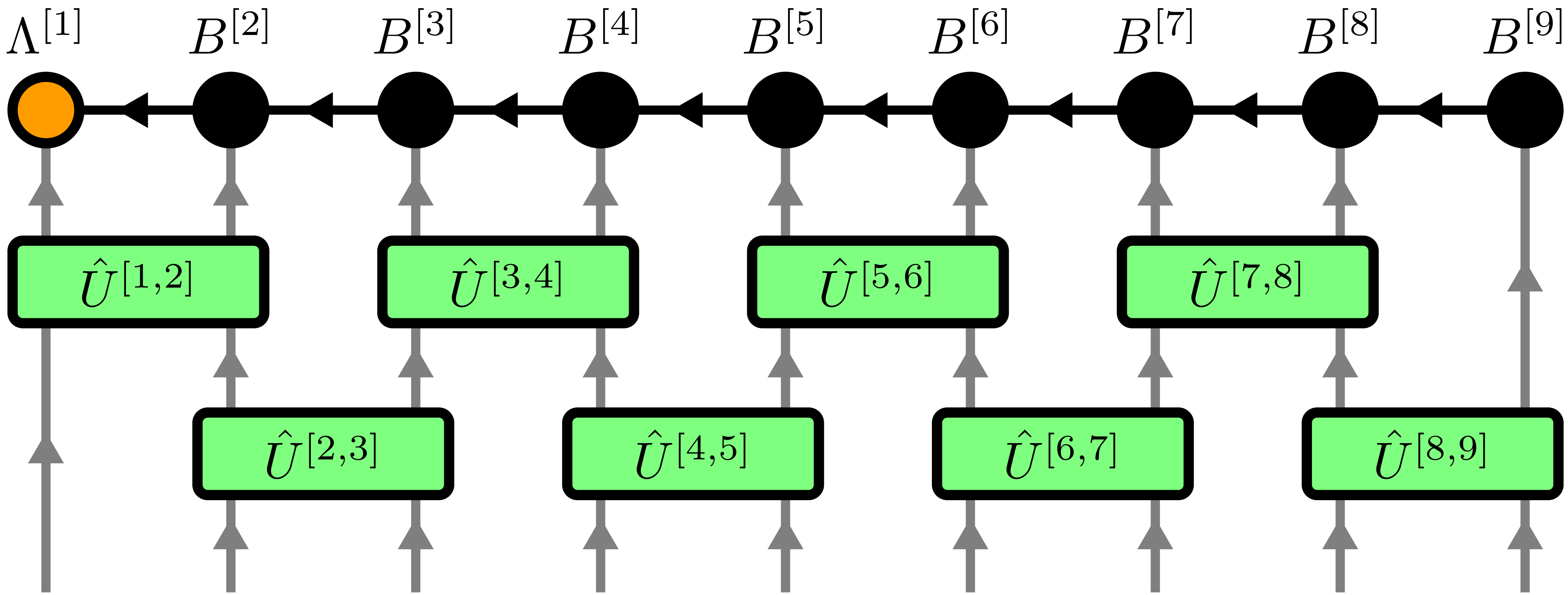
- Hamiltonian $$\hat{H} = \sum_{j\ \text{even}} \hat{h}^{[j,j+1]} + \sum_{j\ \text{odd}} \hat{h}^{[j,j+1]} =: \hat{H}_{\text{even}} + \hat{H}_{\text{odd}}$$

- Suzuki-Trotter decomposition:

$$\hat{U}(\Delta t) = e^{-i\Delta t\left(\hat{H}_{\text{even}}+\hat{H}_{\text{odd}}\right)} = e^{-i\Delta t\hat{H}_{\text{even}}}e^{-i\Delta t\hat{H}_{\text{odd}}} + \mathcal{O}\left(\Delta t^2\right)$$

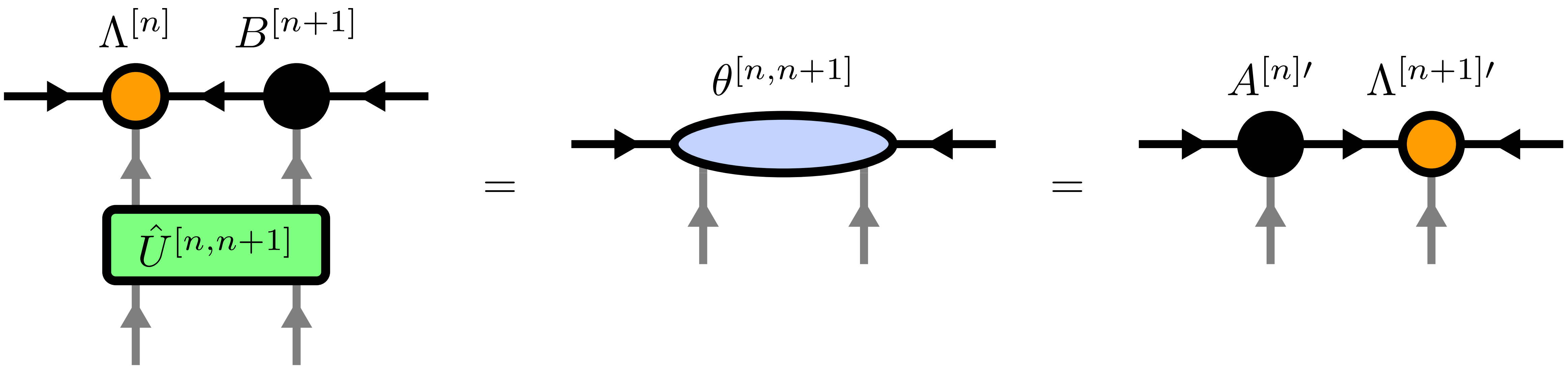
$$e^{-i\Delta t\hat{H}_{\text{even}}} = \prod_{j\ \text{even}} e^{-i\Delta t\hat{h}^{[j,j+1]}} = \prod_{j\ \text{even}} \hat{U}^{[j,j+1]}(\Delta t)$$

Time Evolving Block Decimation (TEBD)



[G. Vidal, 2004]

Time Evolving Block Decimation (TEBD)



[G. Vidal, 2004]

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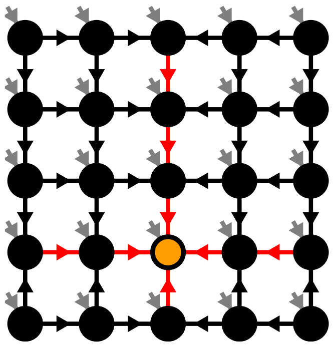
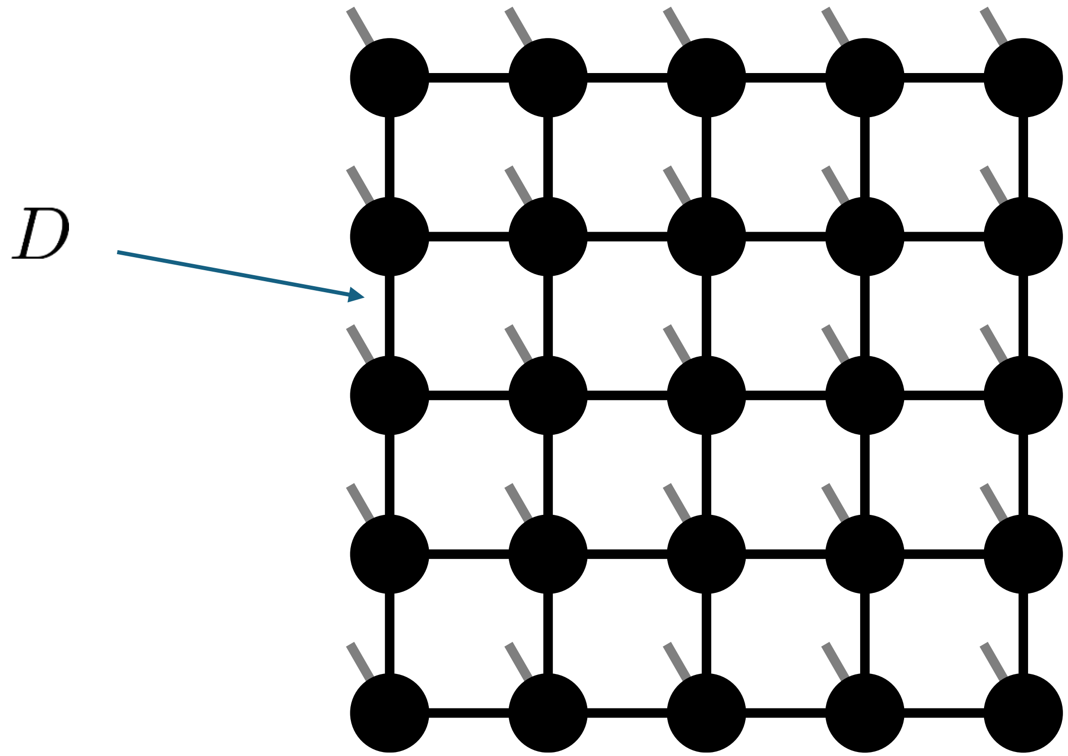


2. Isometric Tensor Product States (isoTPS)

   [M.P. Zalatel, F. Pollmann, 2020]

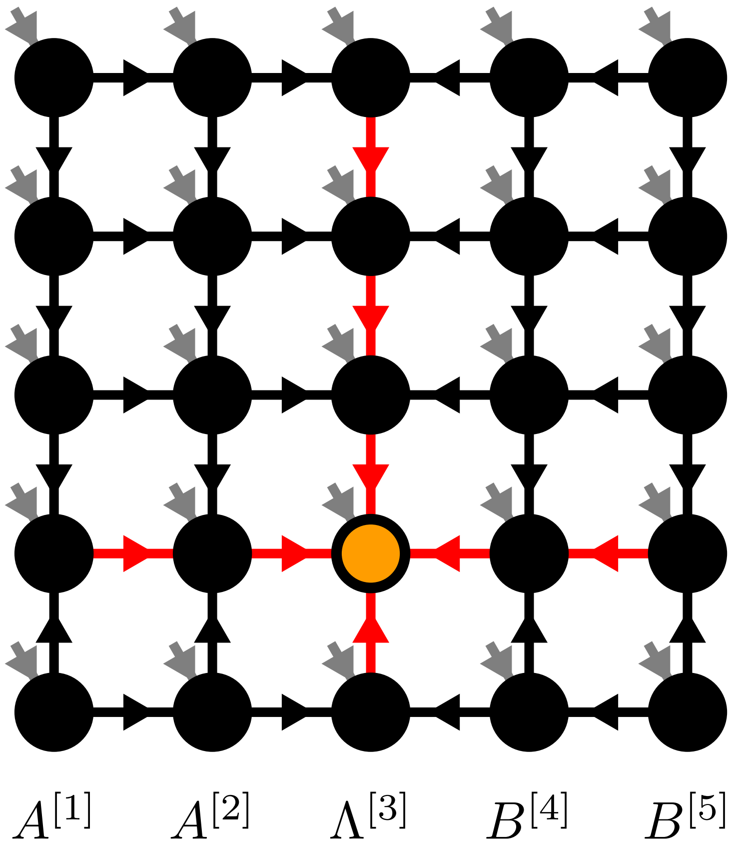
3. Alternative canonical form

Projected Entangled Pair States (PEPS)



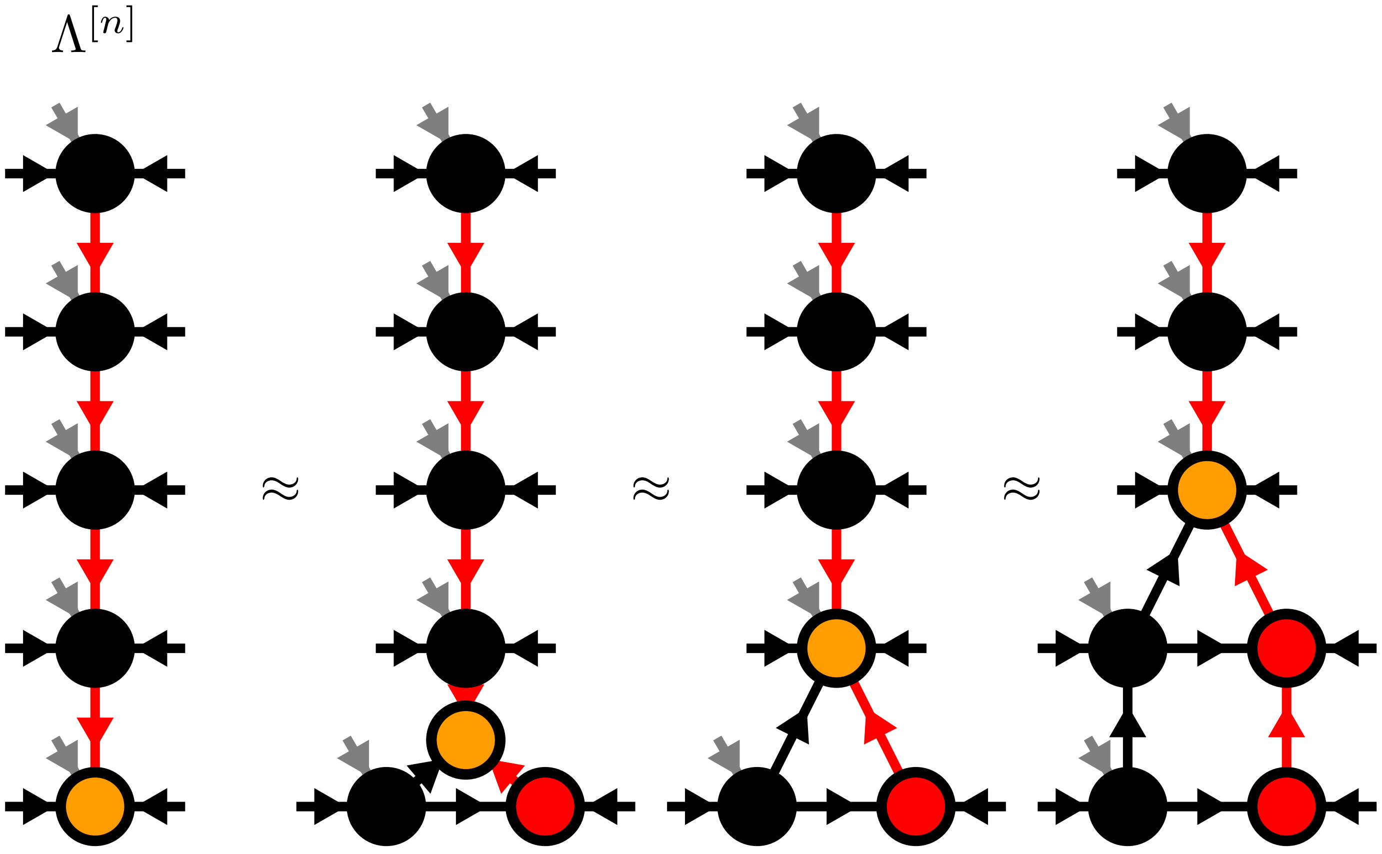
$$\Psi_{i_1,i_2,\cdots,i_N} := \mathcal{C}\left(T^{[1],i_1}, T^{[2],i_2}, \ldots, T^{[N],i_N}\right)$$

Isometric Tensor Product State



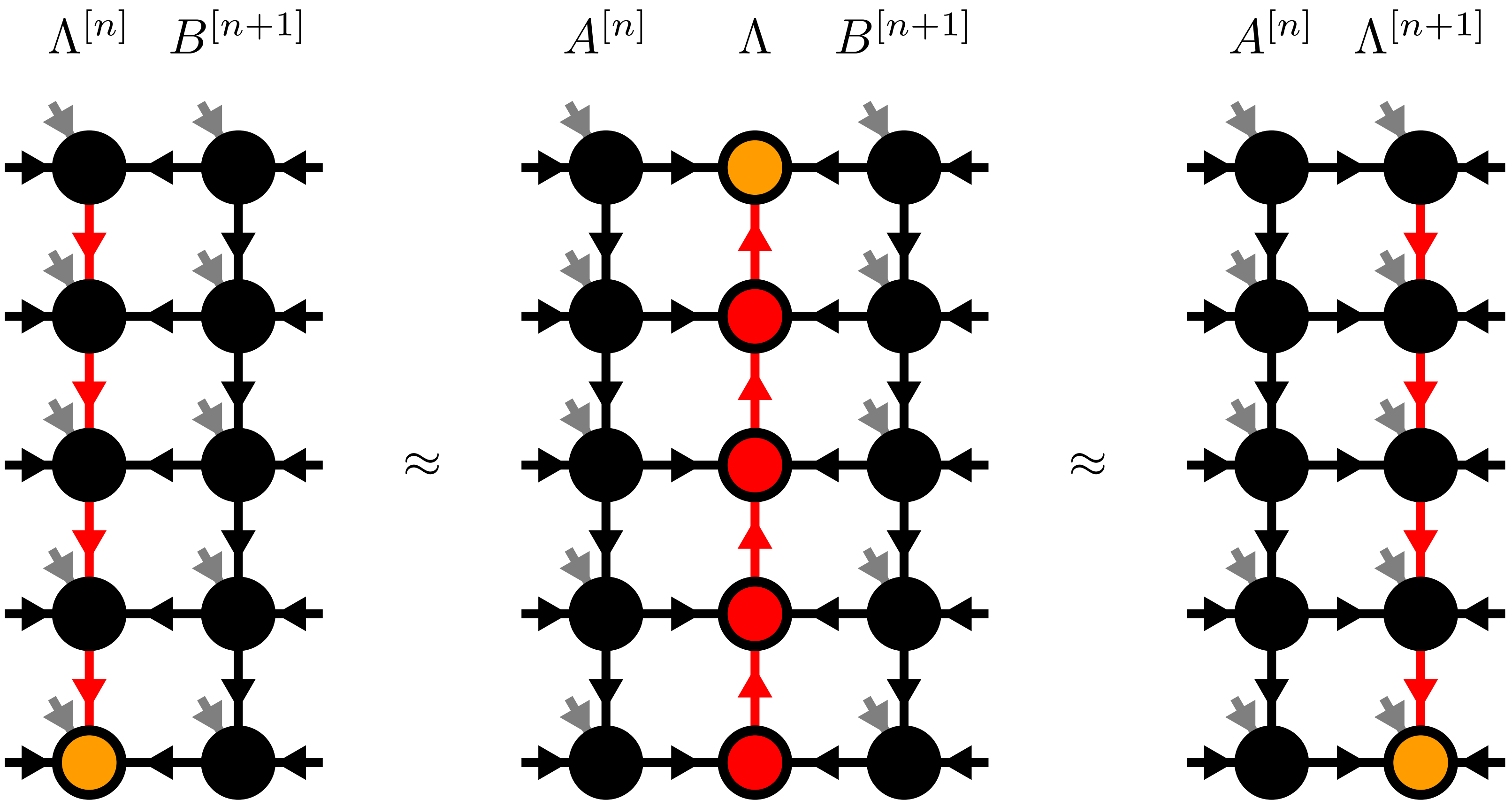
[M.P. Zalatel, F. Pollmann, 2020]

Moses-Move (MM)



[M.P. Zalatel, F. Pollmann, 2020]

Moses-Move (MM)



[M.P. Zalatel, F. Pollmann, 2020]

Algorithms on isoTPS

- TEBD: $\mathcal{O}(D^7)$ [M.P. Zalatel, F. Pollmann, 2020]
- DMRG: $\mathcal{O}(D^{12})$ [S.-H. Lin M.P. Zalatel, F. Pollmann, 2022]
  - generalized eigenvalue problem -> standard eigenvalue problem

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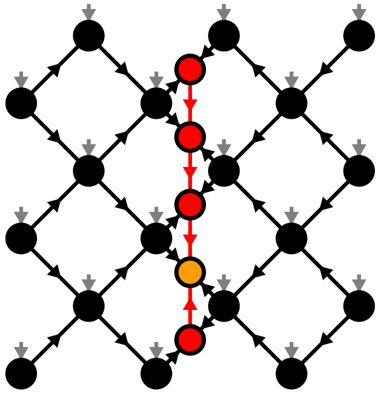
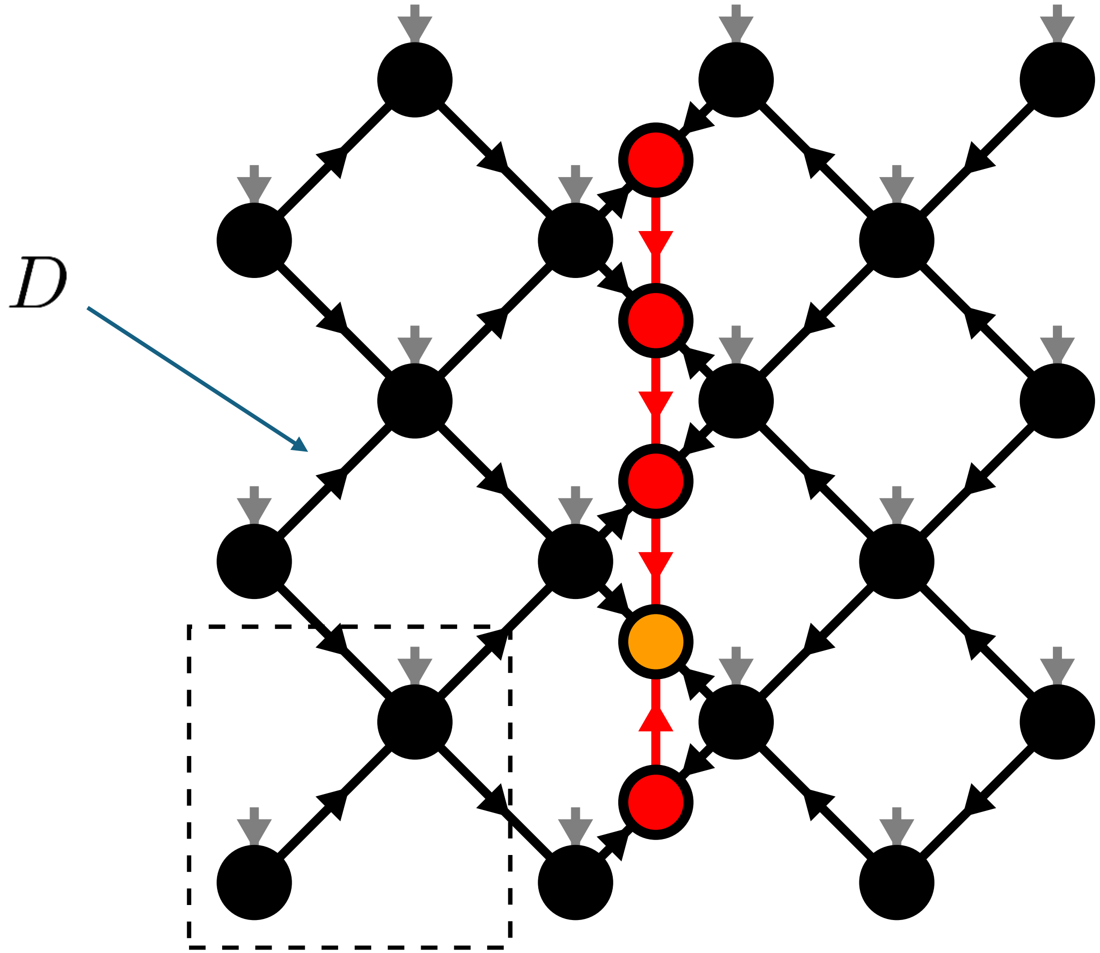
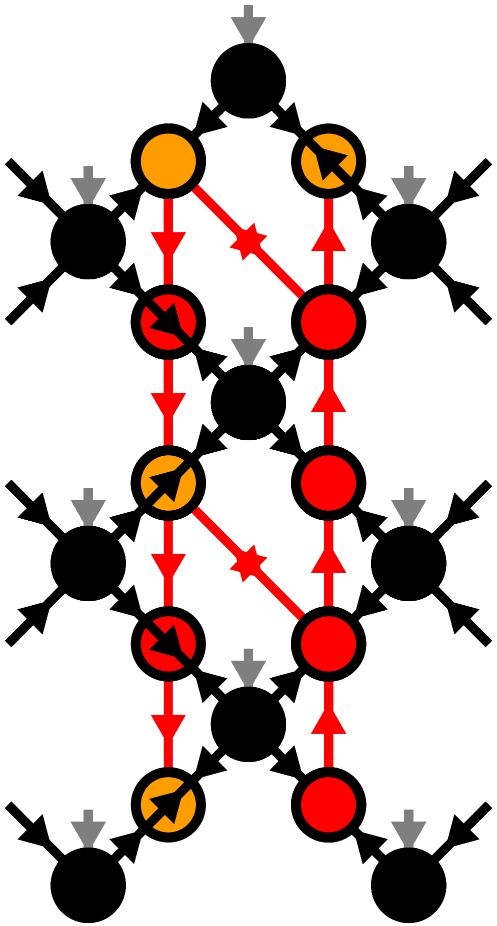


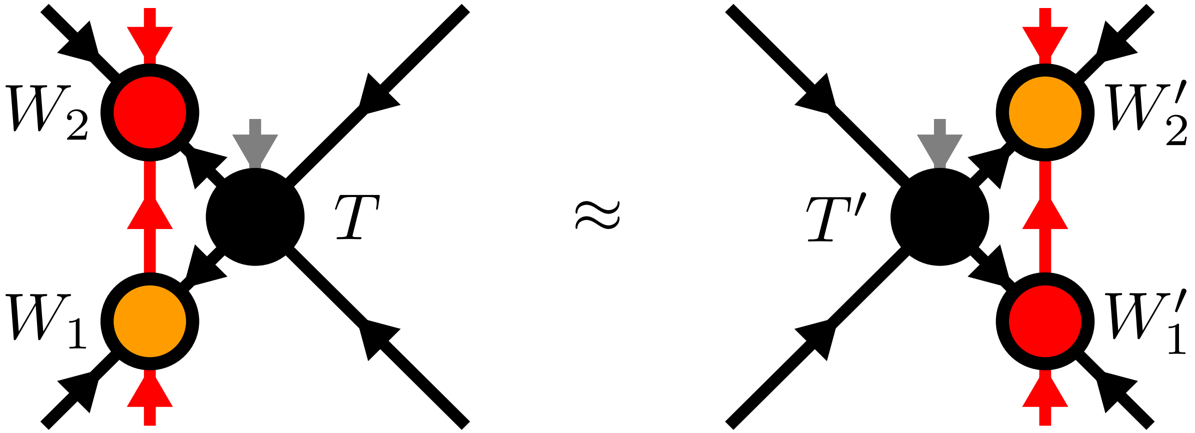
Alternative canonical form



Moving the orthogonality hypersurface



Yang-Baxter (YB) Move



- YB Error $\quad \| |\Psi\rangle - |\Psi'\rangle \| = \sqrt{2 - 2\,\mathrm{Re}\,\langle \Psi' | \Psi \rangle}$

Yang-Baxter (YB) Move

- Constrained optimization problem

$$(T'_{\mathrm{opt}}, W'_{1,\mathrm{opt}}, W'_{2,\mathrm{opt}}) = \underset{T, W'_1, W'_2}{\operatorname{argmax}} \operatorname{Re} \langle \Psi' | \Psi \rangle$$

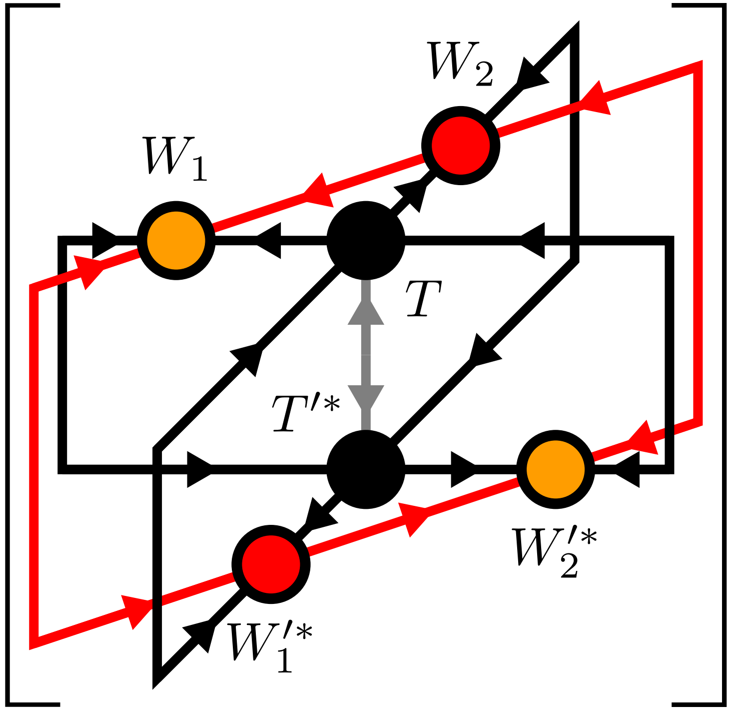
$$T'^{\dagger} T' = \mathbb{1}, \quad W_1'^{\dagger} W_1' = \mathbb{1}, \quad \| W_2' \|_{\mathrm{F}} = 1$$

- Two algorithms:
  - Variational optimization with local updates
  - Tripartite decomposition using SVD + disentangling

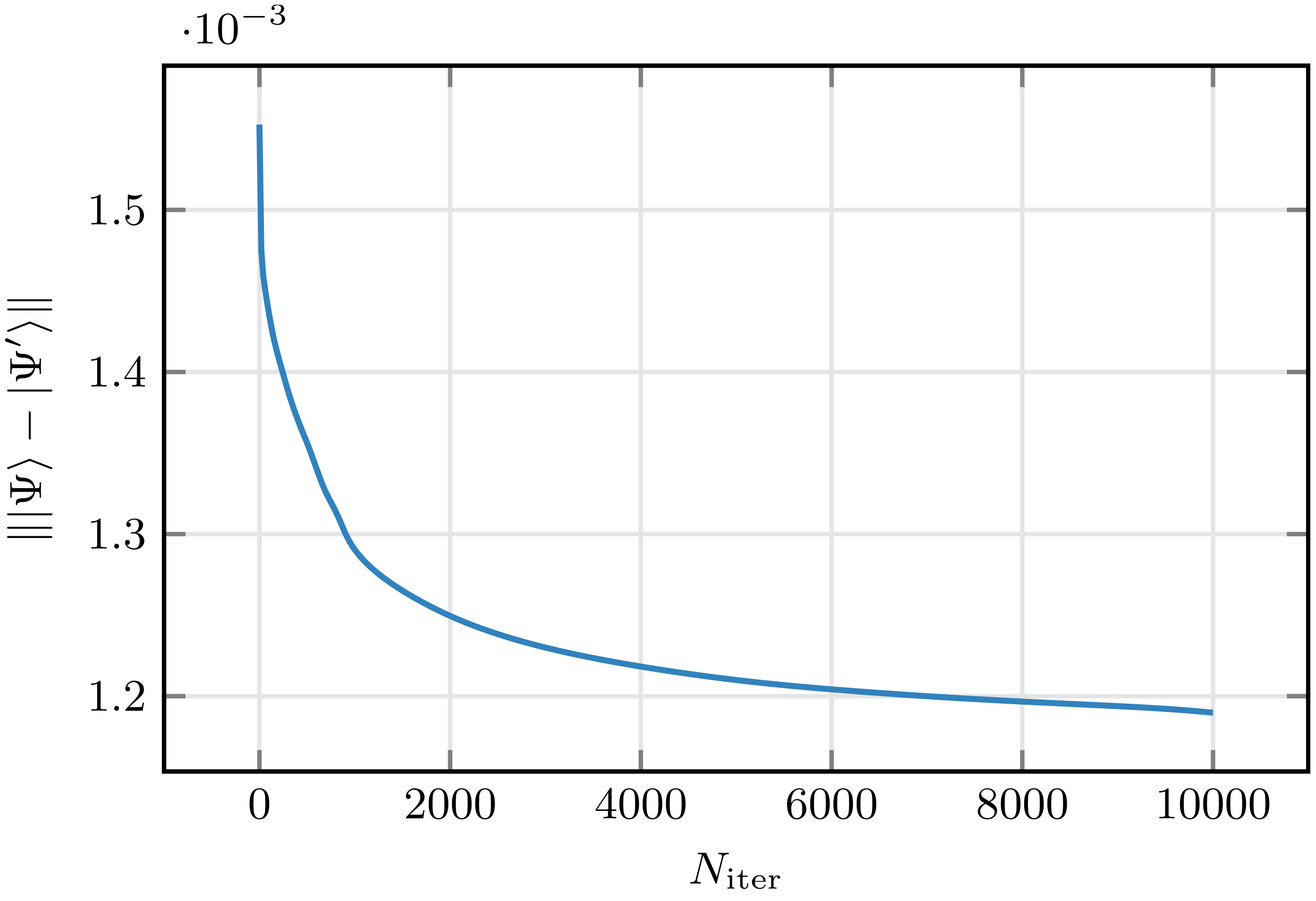
YB-move: Variational optimization

$$(T'_{\rm opt}, W'_{1,\rm opt}, W'_{2,\rm opt}) = \underset{T, W'_1, W'_2}{\rm argmax}\, {\rm Re}\, \langle \Psi' | \Psi \rangle$$

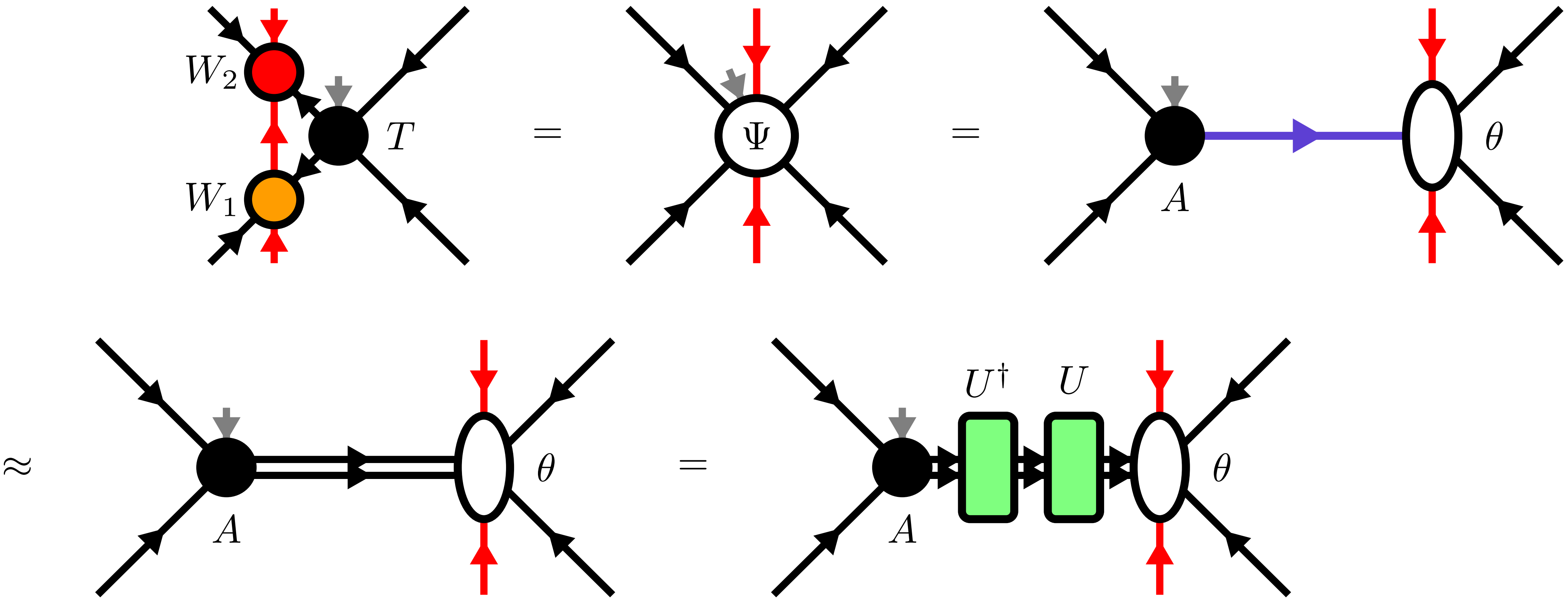
$${\rm Re}\, \langle \Psi' | \Psi \rangle = {\rm Re}$$



YB-move: Variational optimization

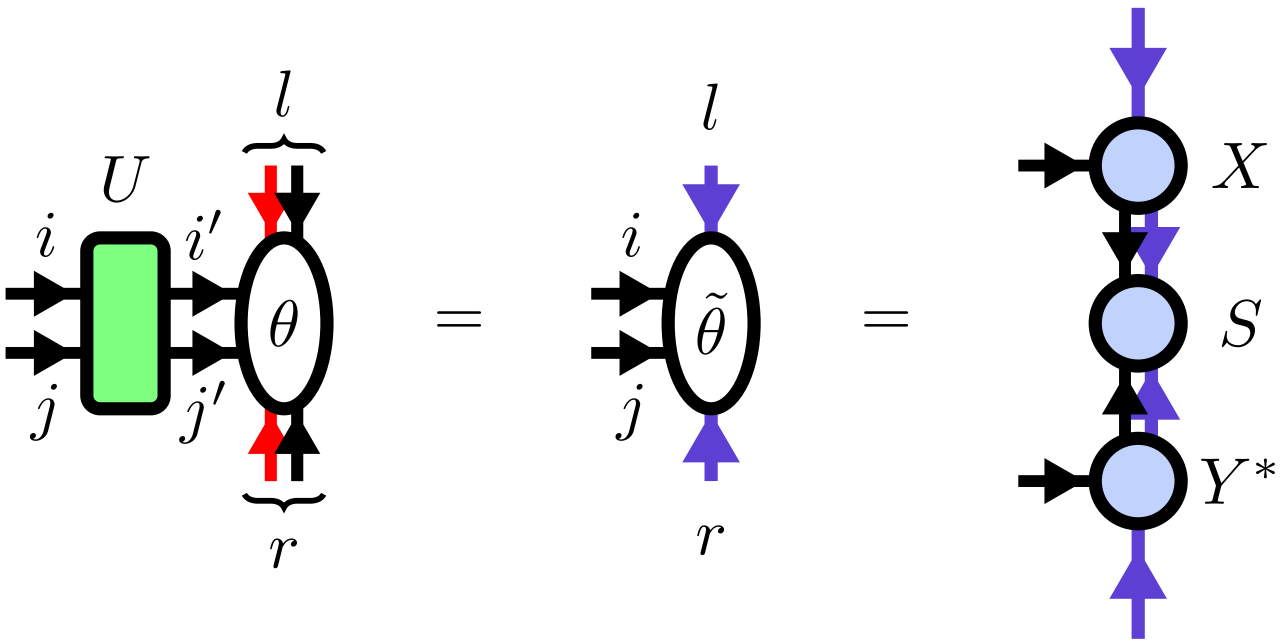


YB-move: Tripartite decomposition



[M.P. Zalatel, F. Pollmann, 2020]

Disentangling procedure

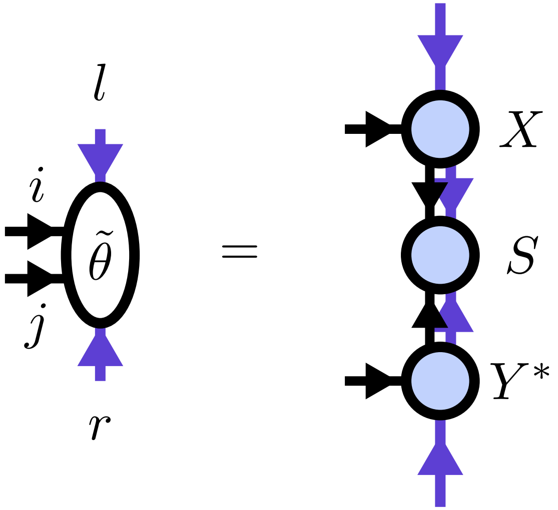


Disentangling procedure

- Minimize the *Rényi-entropy*

$$f_{\text{Rényi}}\left(U, \theta, \alpha\right) = \frac{1}{1-\alpha} \log \operatorname{Tr}\left(\rho^{\alpha}\right)$$

$$\rho = \operatorname{Tr}_{(j,r)}(|\tilde{\theta}\rangle \langle\tilde{\theta}|)$$



- Bounds truncation error for $\alpha \leq 1$
- In practice: Good results also for $\alpha > 1$

[M.P. Zalatel, F. Pollmann, 2020]

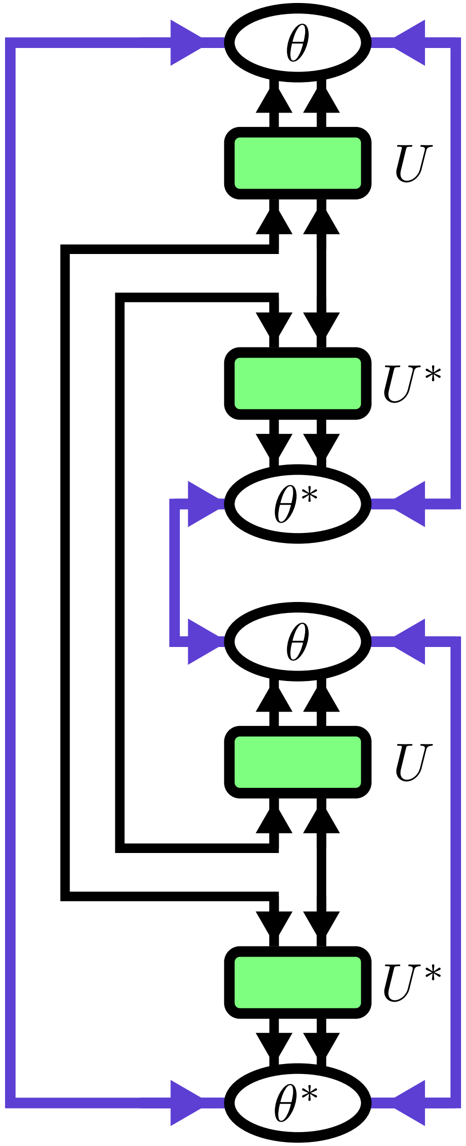
Renyi-2 disentangling

- Cost function:

$$f_{\text{Rényi}}\left(U,\theta,\alpha=2\right) = -\log \operatorname{Tr}\rho^2$$

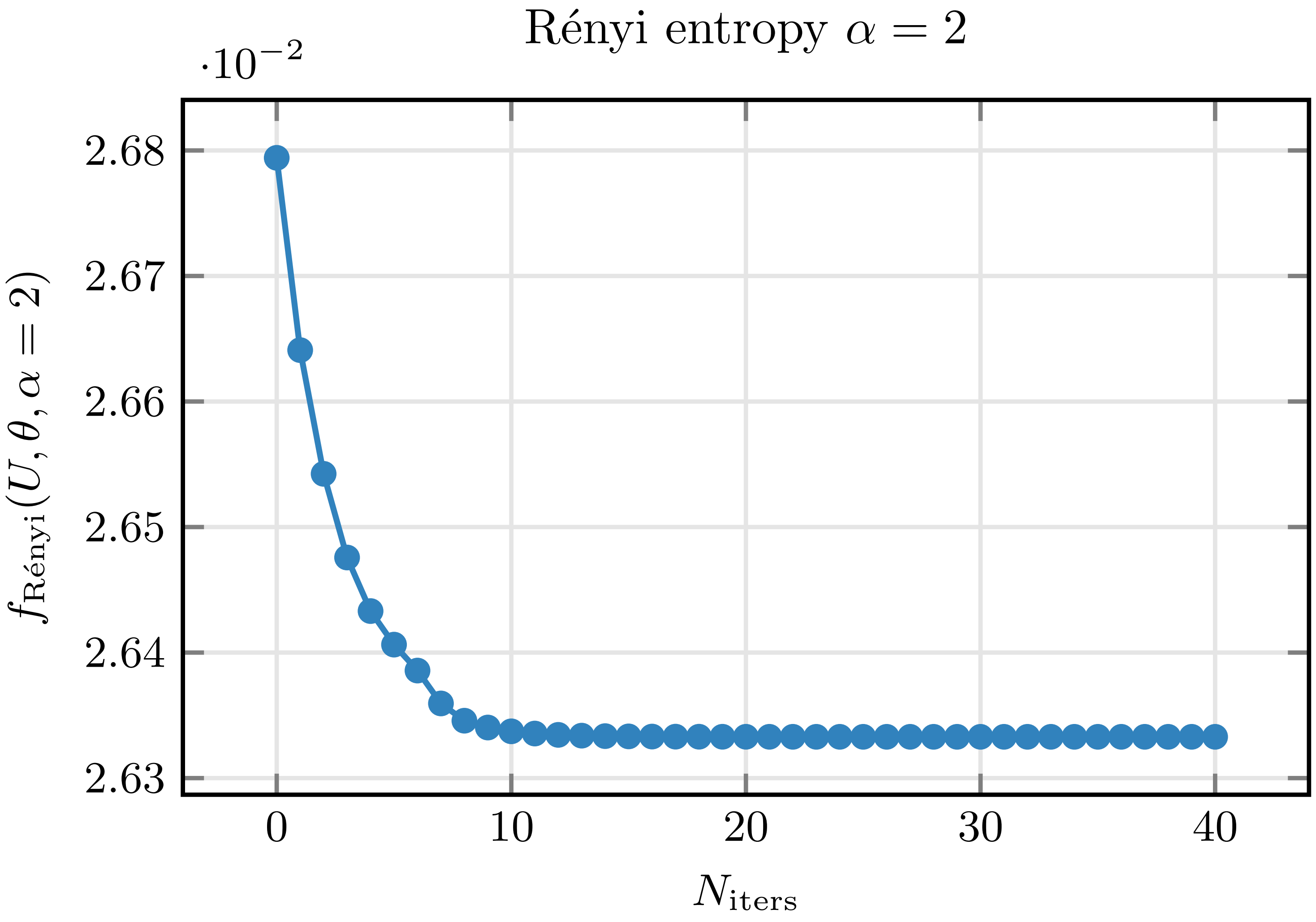
$$\Rightarrow \quad \text{maximize} \quad \operatorname{Tr}\rho^2$$

- Evenbly-Vidal algorithm: $\mathcal{O}(D^9)$



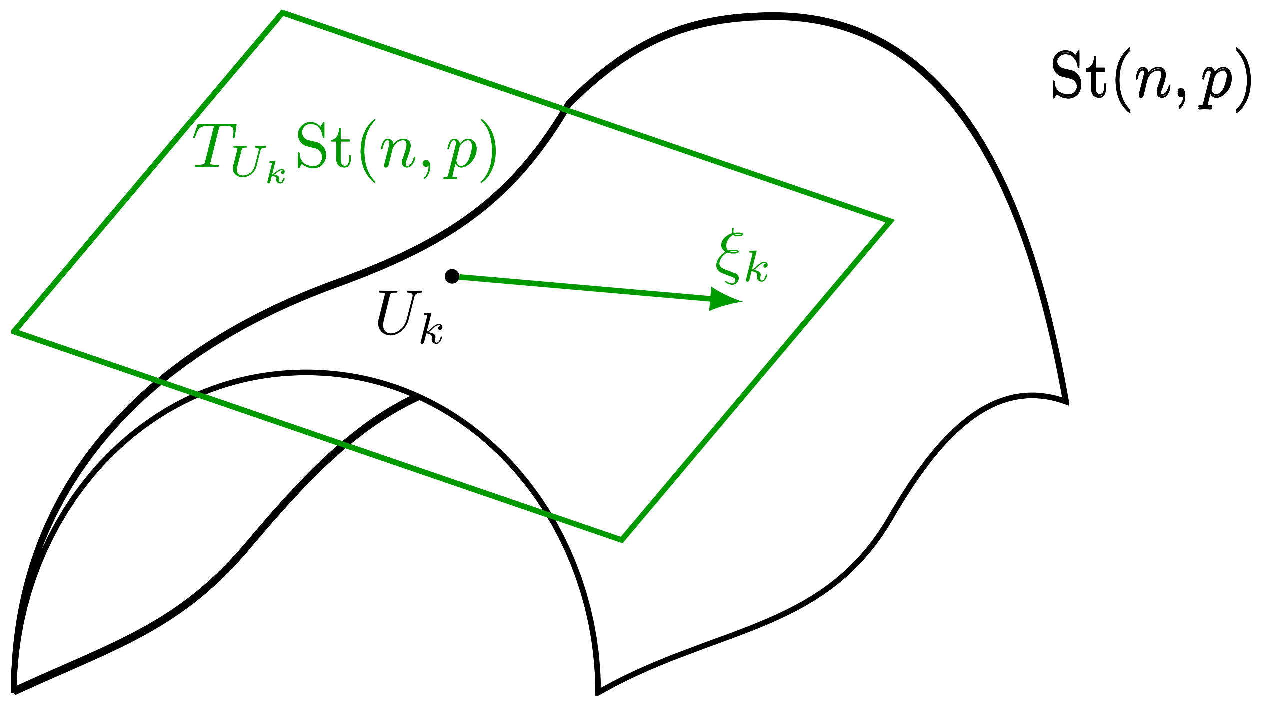
[J. Hauschild et al. 2018]

Renyi-2 disentangling



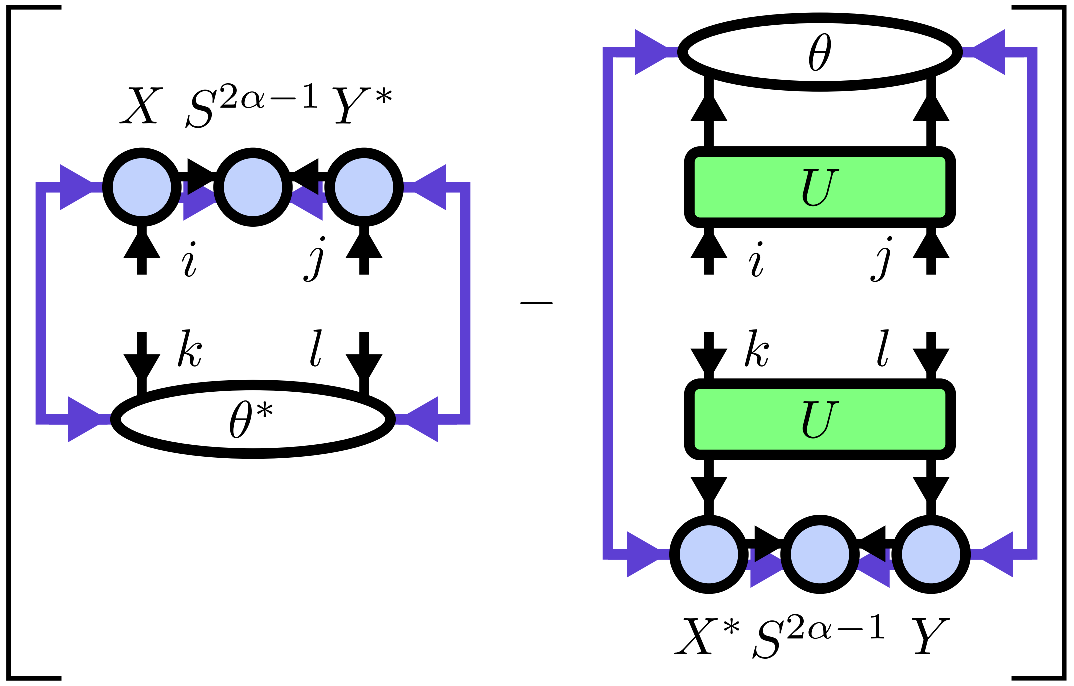
Riemannian optimization

- No easy tensor network form for $\alpha \neq 2$
- Use Riemannian optimization! [S.T. Smith, 2014]



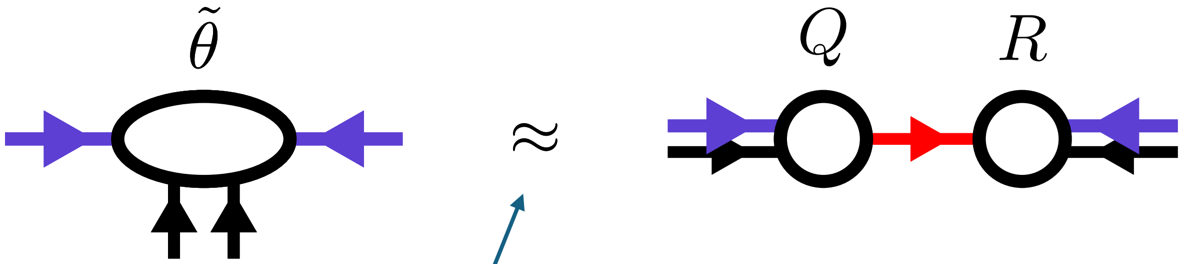
Riemannian optimization

$$\left(\nabla f_{\text{Rényi}} \left(U, \theta, \alpha \right) \right)_{i,j,k,l} \propto$$

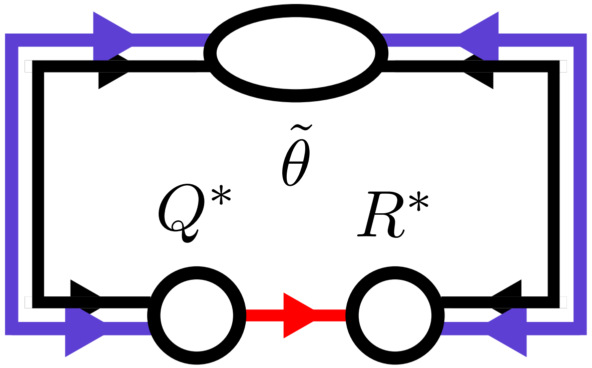


- Computing Gradient costs $\mathcal{O}(D^9)$

Approximate Gradients



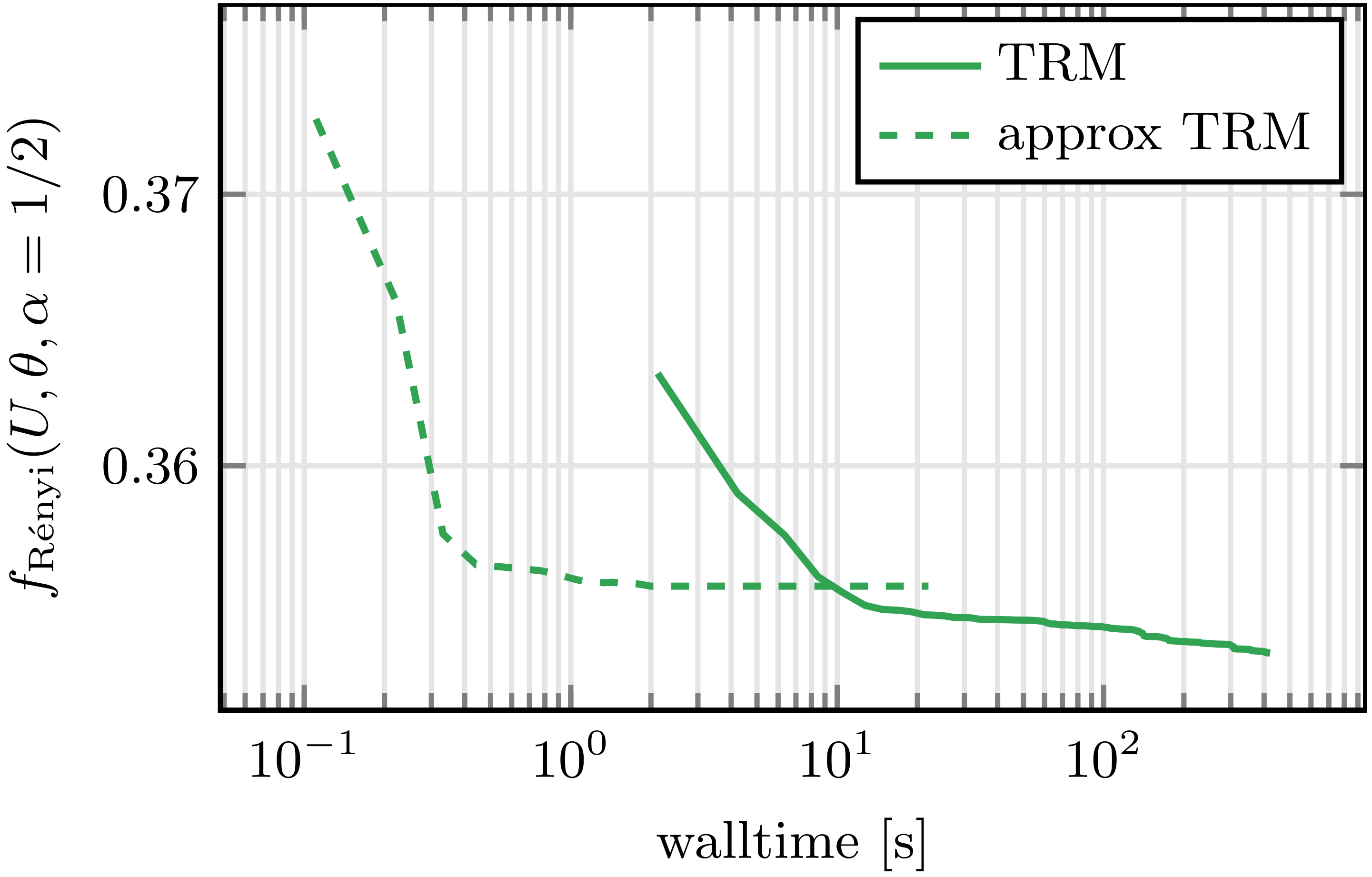
$\langle \tilde{\theta}, QR \rangle_{\mathrm{F}} \quad =$



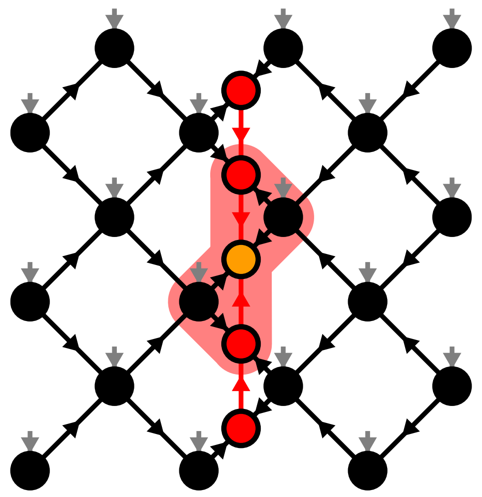
$$\mathcal{O}(D^9) \to \mathcal{O}(D^8)$$

[J. Unfried, J. Hauschild, F. Pollmann, 2022]

Approximate Gradients



TEBD: Local updates



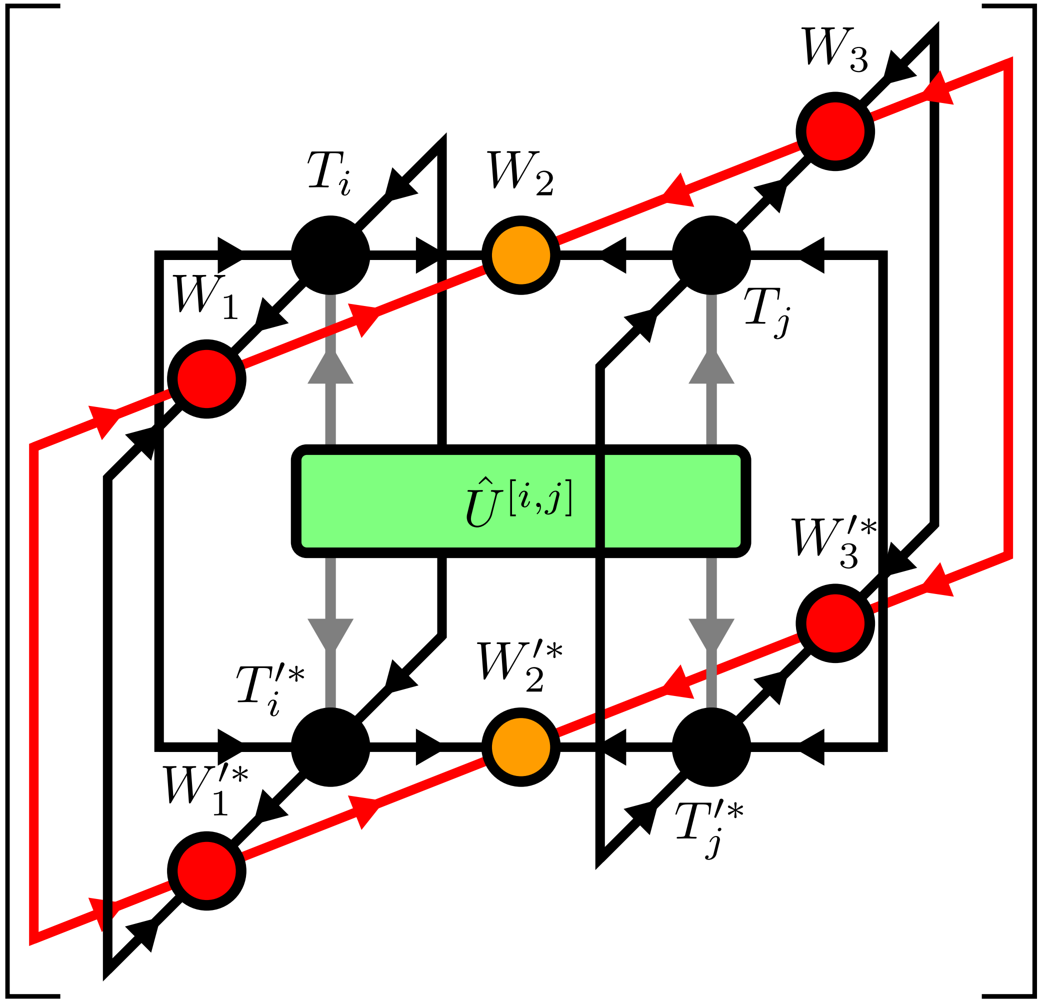
- Apply local bond operator: $|\Psi'\rangle = \hat{U}^{[i,j]}(\Delta t)\,|\Psi\rangle$
- Optimization problem:

$$(T'_{i,\mathrm{opt}}, T'_{j,\mathrm{opt}}, W'_{1,\mathrm{opt}}, W'_{2,\mathrm{opt}}, W'_{3,\mathrm{opt}}) =$$

$$\operatorname*{argmax}_{T'_i, T'_j, W'_1, W'_2, W'_3} \mathrm{Re}\,\langle\Psi'|\,\hat{U}^{[i,j]}(\Delta t)\,|\Psi\rangle$$

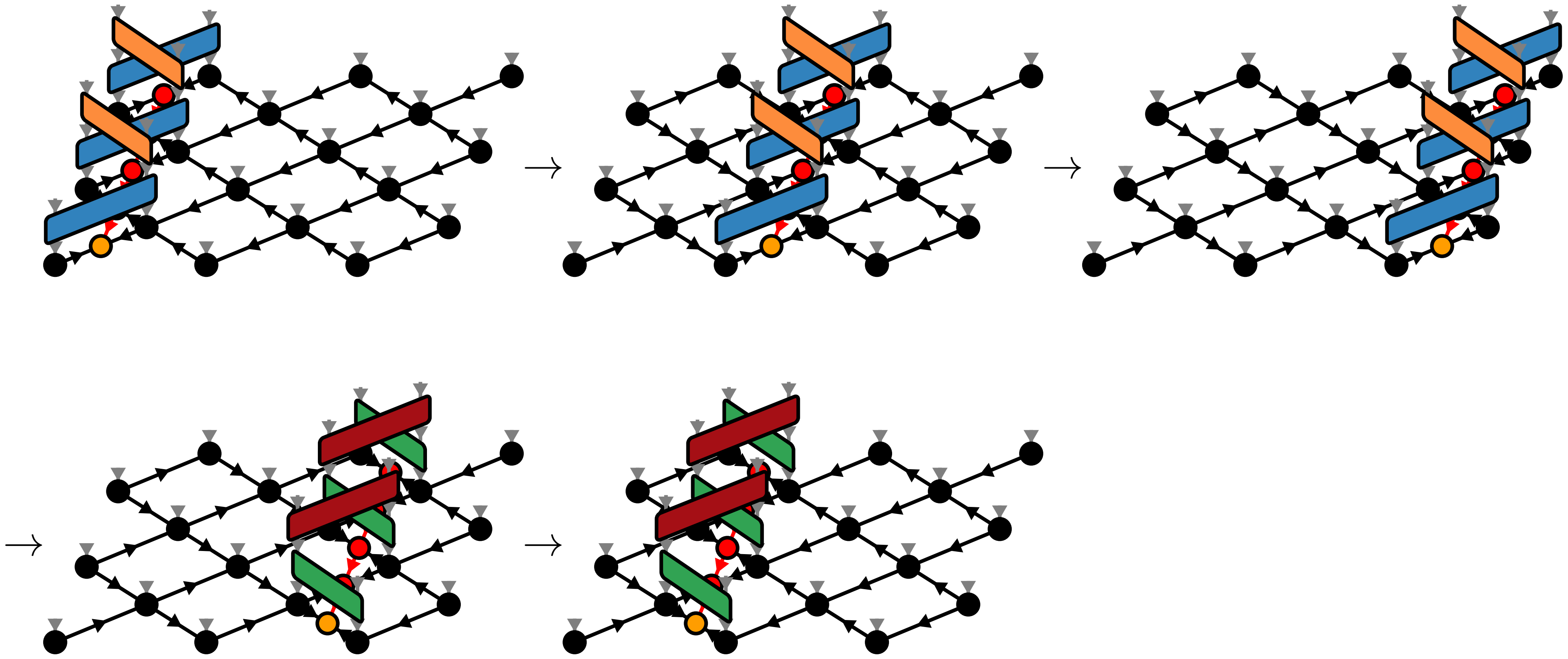
TEBD: Local updates

$$\mathrm{Re}\,\langle\Psi'|\hat{U}^{[i,j]}(\Delta t)|\Psi\rangle = \mathrm{Re}$$



TEBD1

$$\hat{U}(\Delta t) = \hat{U}^{\mathrm{TEBD1}}(\Delta t) + \mathcal{O}(\Delta t^2)$$



TEBD1 vs TEBD2

- First order: $\hat{U}(\Delta t) = \hat{U}^{\mathrm{TEBD1}}(\Delta t) + \mathcal{O}(\Delta t^2)$

- Second order: $\hat{U}(\Delta t) = \hat{U}^{\mathrm{TEBD2}}(\Delta t) + \mathcal{O}(\Delta t^3)$

with the same number of YB moves!

Transverse Field Ising (TFI) model

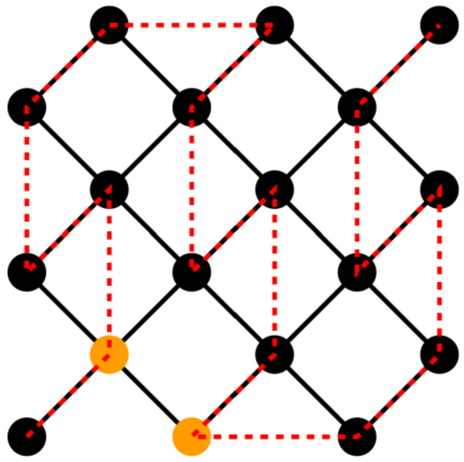
$$\hat{H}_{\mathrm{TFI}} = -J \sum_{\langle i,j \rangle} \hat{\sigma}_i^x \hat{\sigma}_j^x - g \sum_i \hat{\sigma}_i^z$$

- Critical field $g \approx 3.04438$ [H.W. J. Blöte, Y. Deng, 2002]

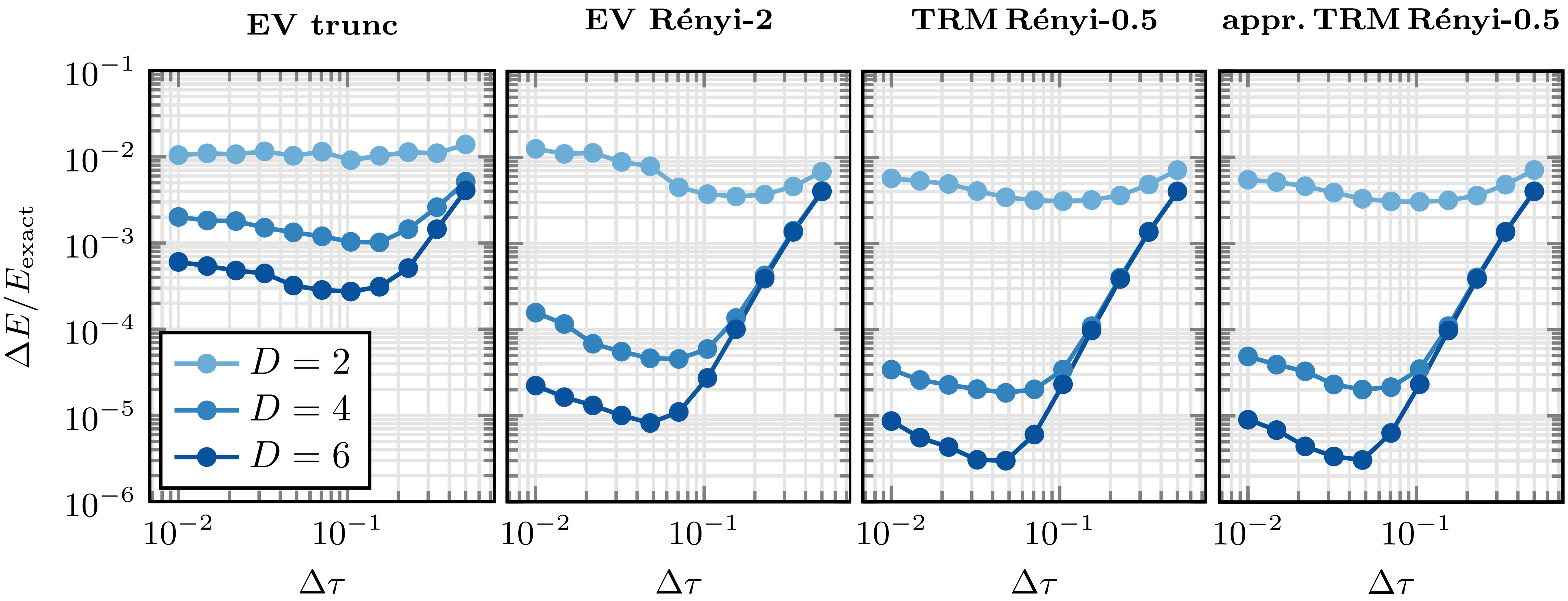
Ground state search with imaginary TEBD

- Imaginary TEBD can be used to find ground states
- TEBD error: $\varepsilon_{\mathrm{TEBD}} = \varepsilon_{\mathrm{Trotter}} + \varepsilon_{\mathrm{local}} + \varepsilon_{\mathrm{YB}}$
- Best results: $\varepsilon_{\mathrm{Trotter}} + \varepsilon_{\mathrm{local}} \approx \varepsilon_{\mathrm{YB}}$
- Reference Simulation: DMRG with tenpy

  [J. Hauschild, F. Pollmann, 2018]

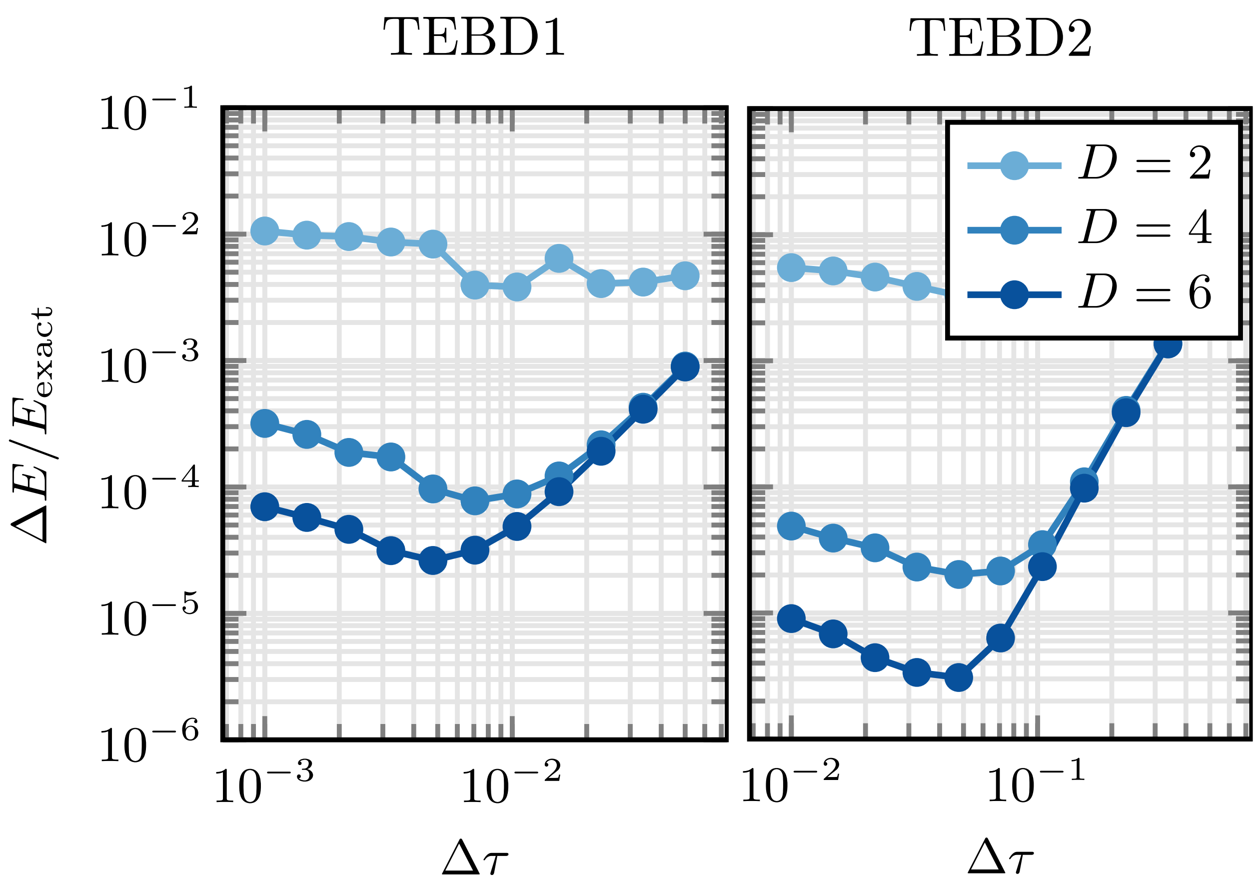


Comparing YB-move algorithms



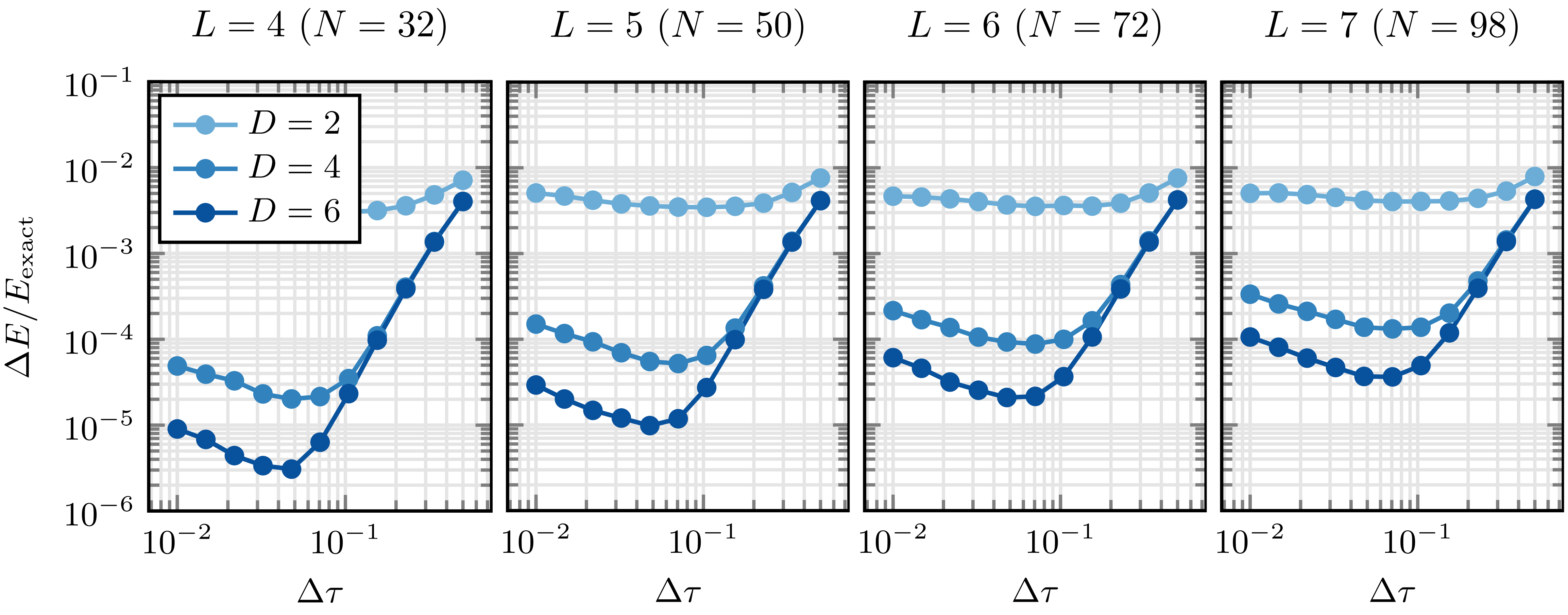
4x4 diagonal square lattice, TFI model with $g/J = 3.5$

TEBD1 vs TEBD2



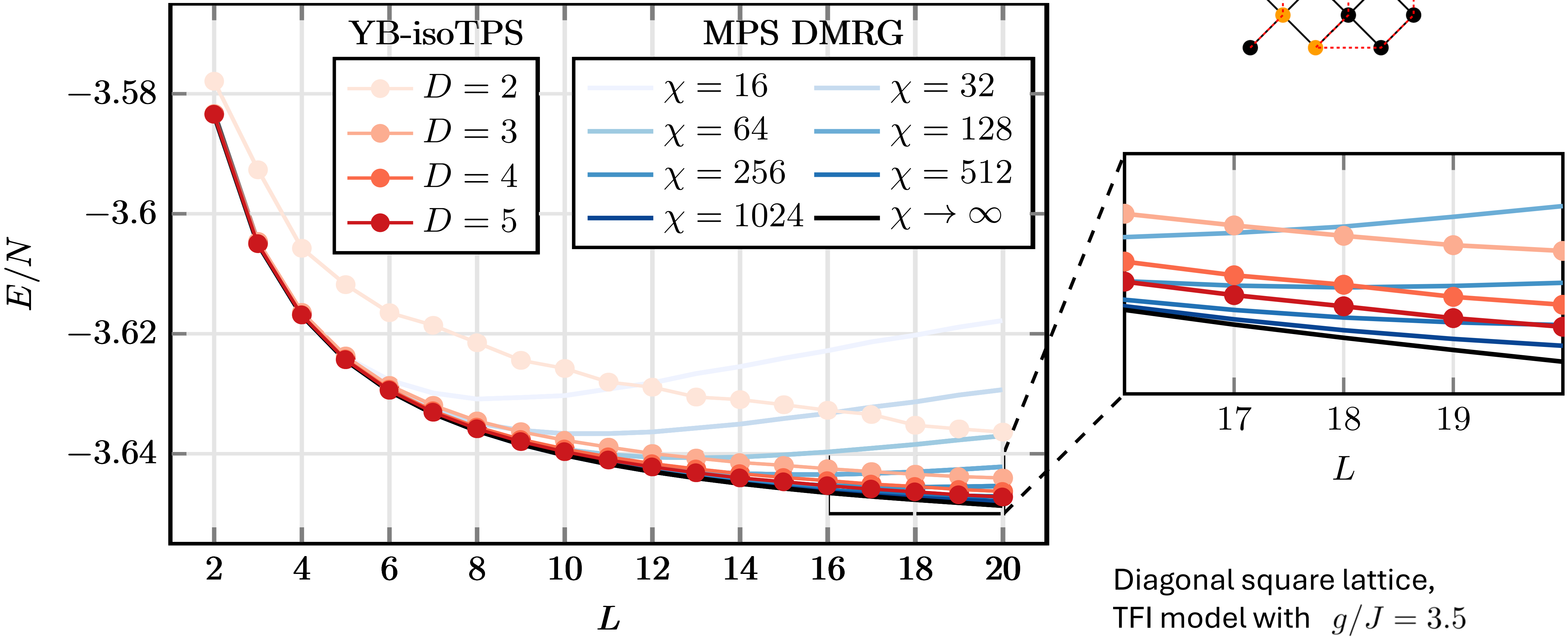
4x4 diagonal square lattice, TFI model with $g/J = 3.5$

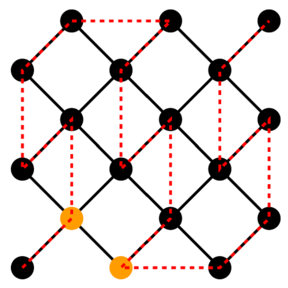
Larger lattices



Diagonal square lattice, TFI model with $g/J = 3.5$

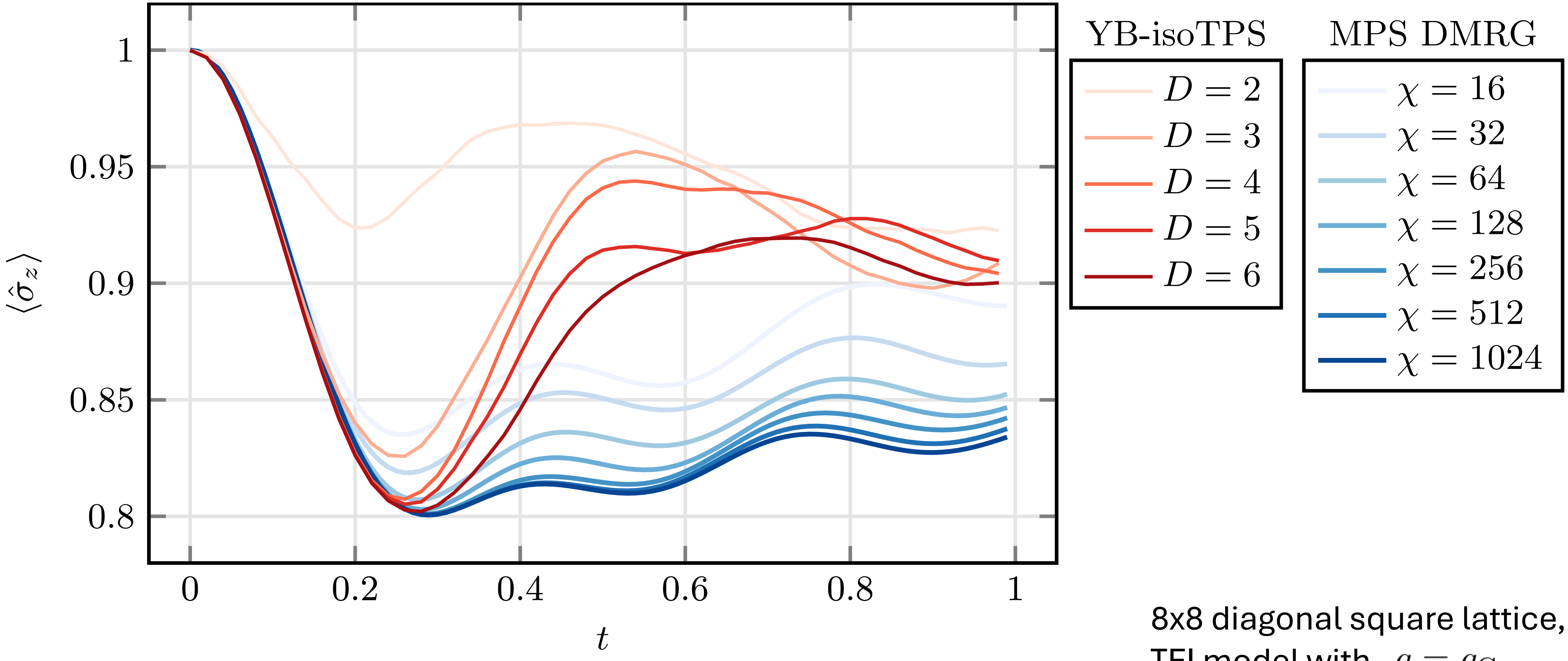
Larger lattices





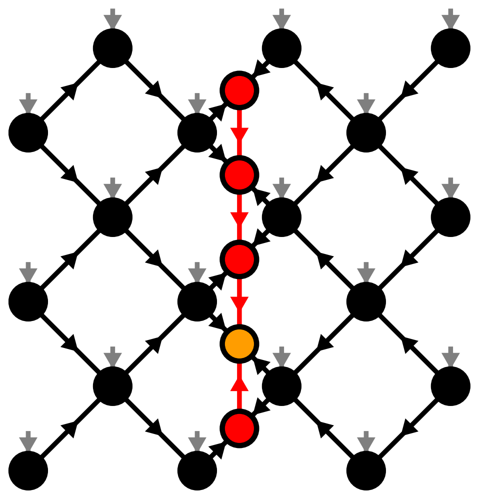
Diagonal square lattice,
TFI model with $g/J = 3.5$

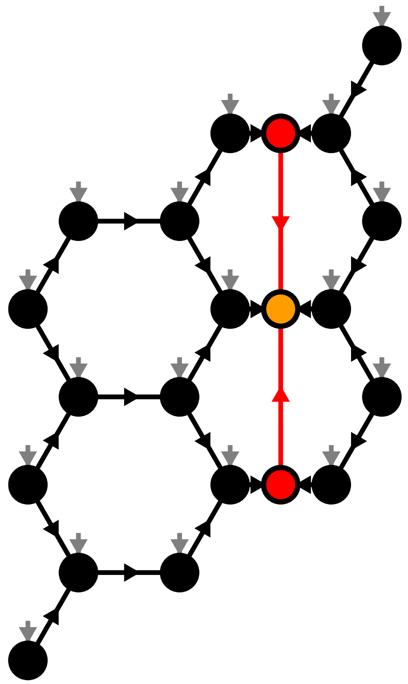
Real-time evolution: Global quench



8x8 diagonal square lattice,
TFI model with $g = g_C$

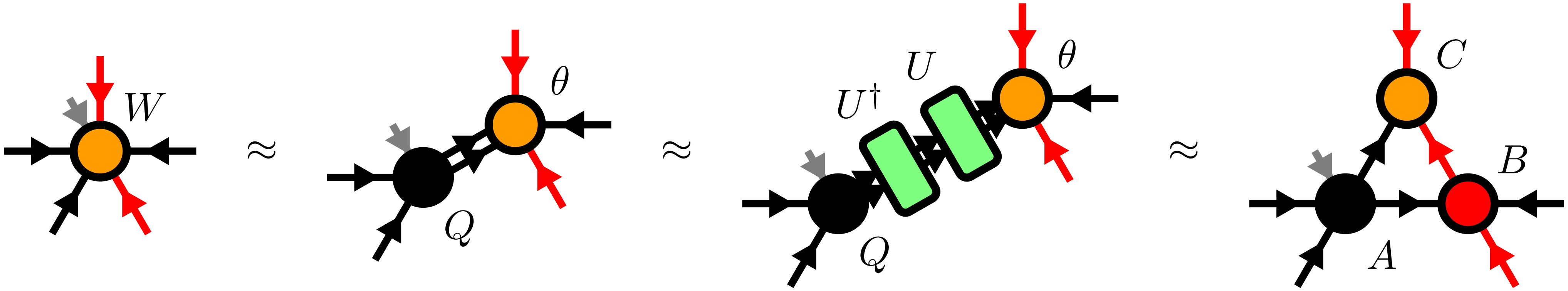
Summary & Outlook



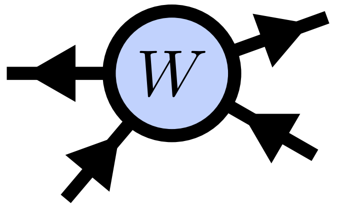


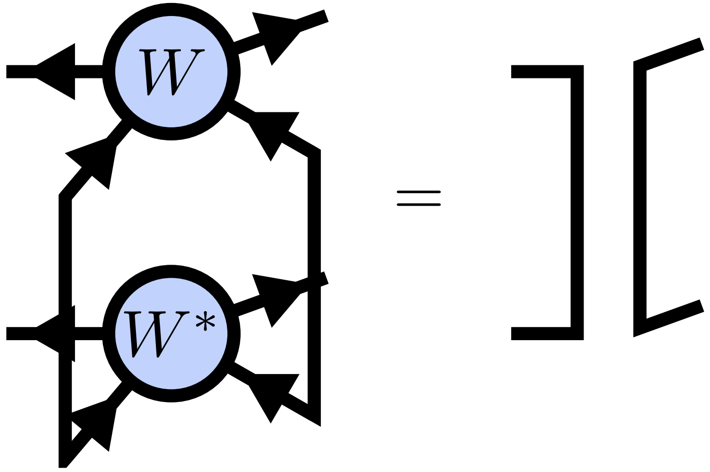
- Alternative canonical form
- YB-move: $\mathcal{O}(D^8)$
- Easily generalizable to other lattices
- Toric Code ground state exactly representable
- TEBD algorithm for time evolution
- Numerical evidence for 2D area law states
- Outlook:
  - DMRG
  - Thermodynamic limit
  - GPUs

Moses-Move (MM) – Tripartite Decomposition

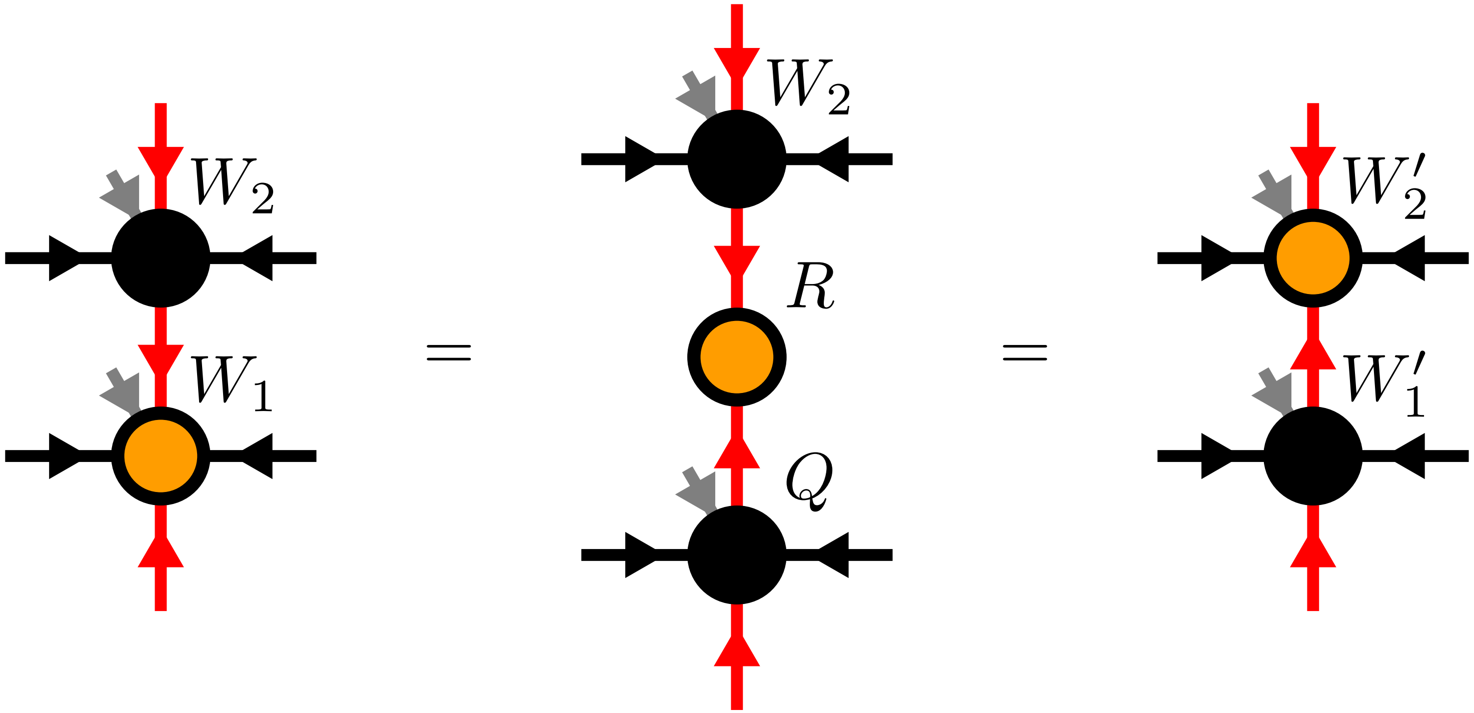


Isometric Tensors

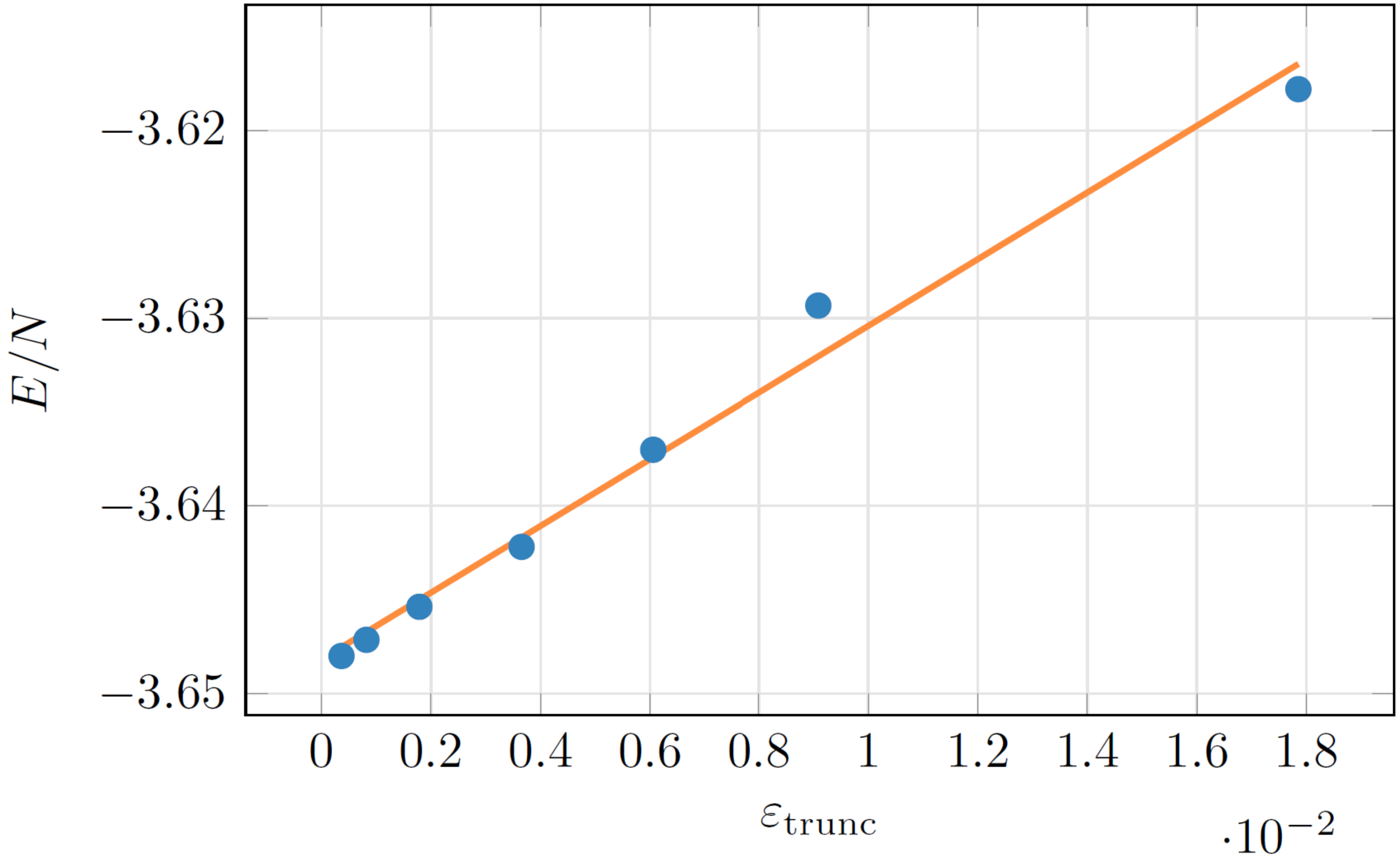




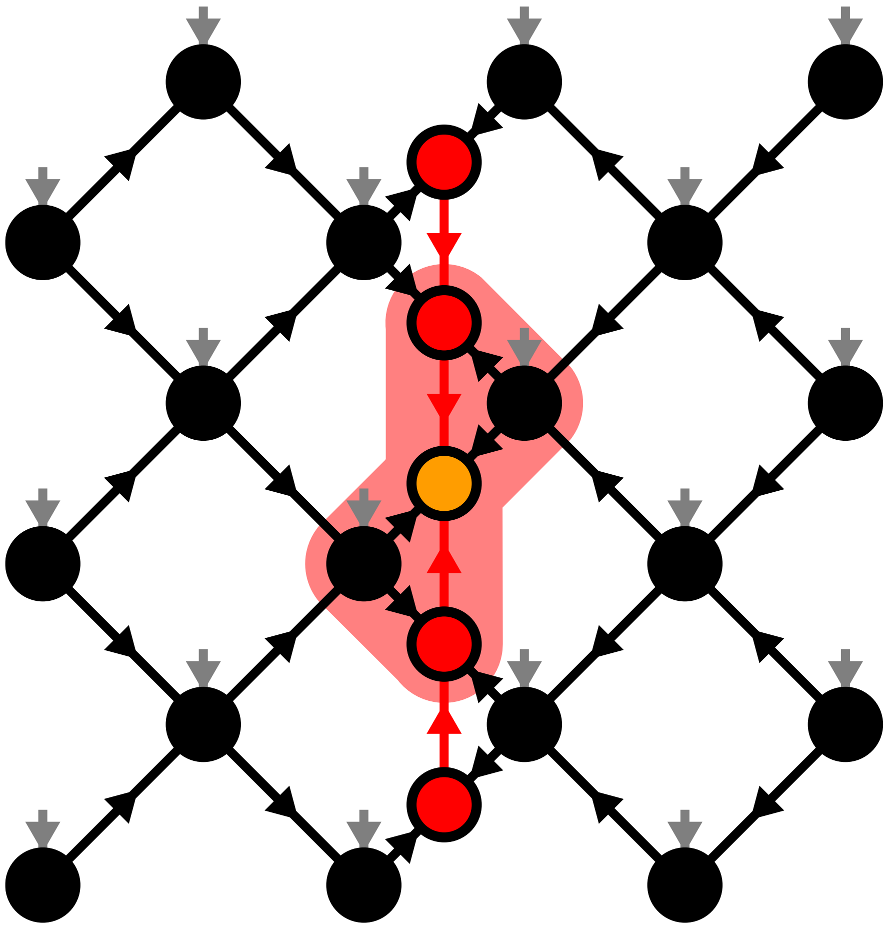
Moving the Orthogonality Center



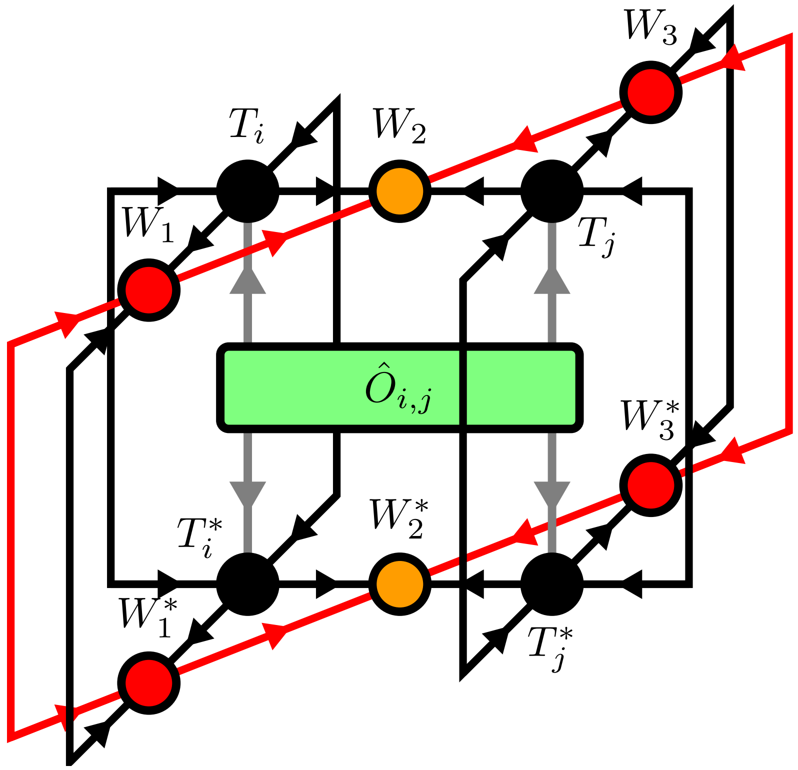
DMRG extrapolation



Local Expectation Values

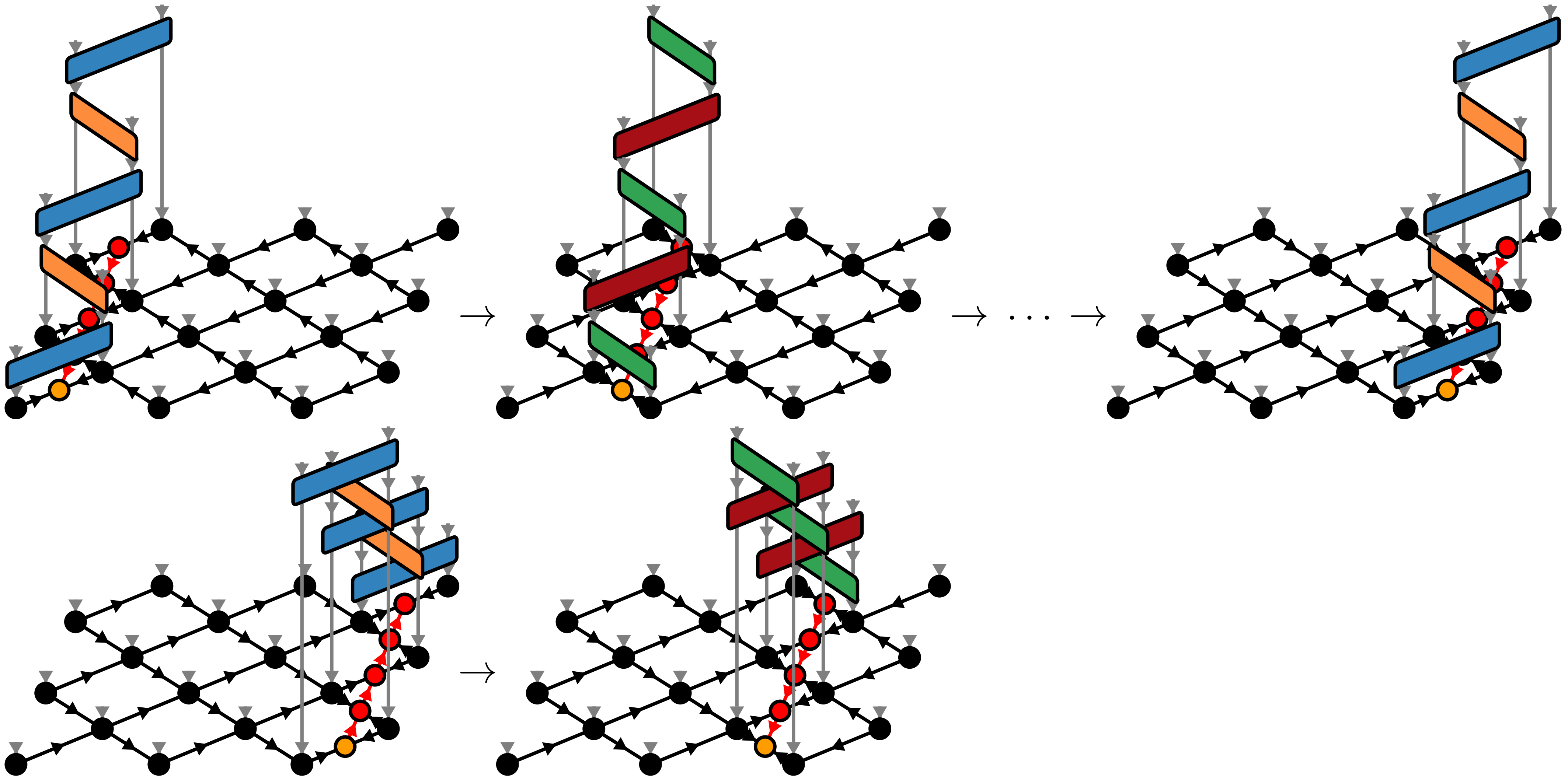


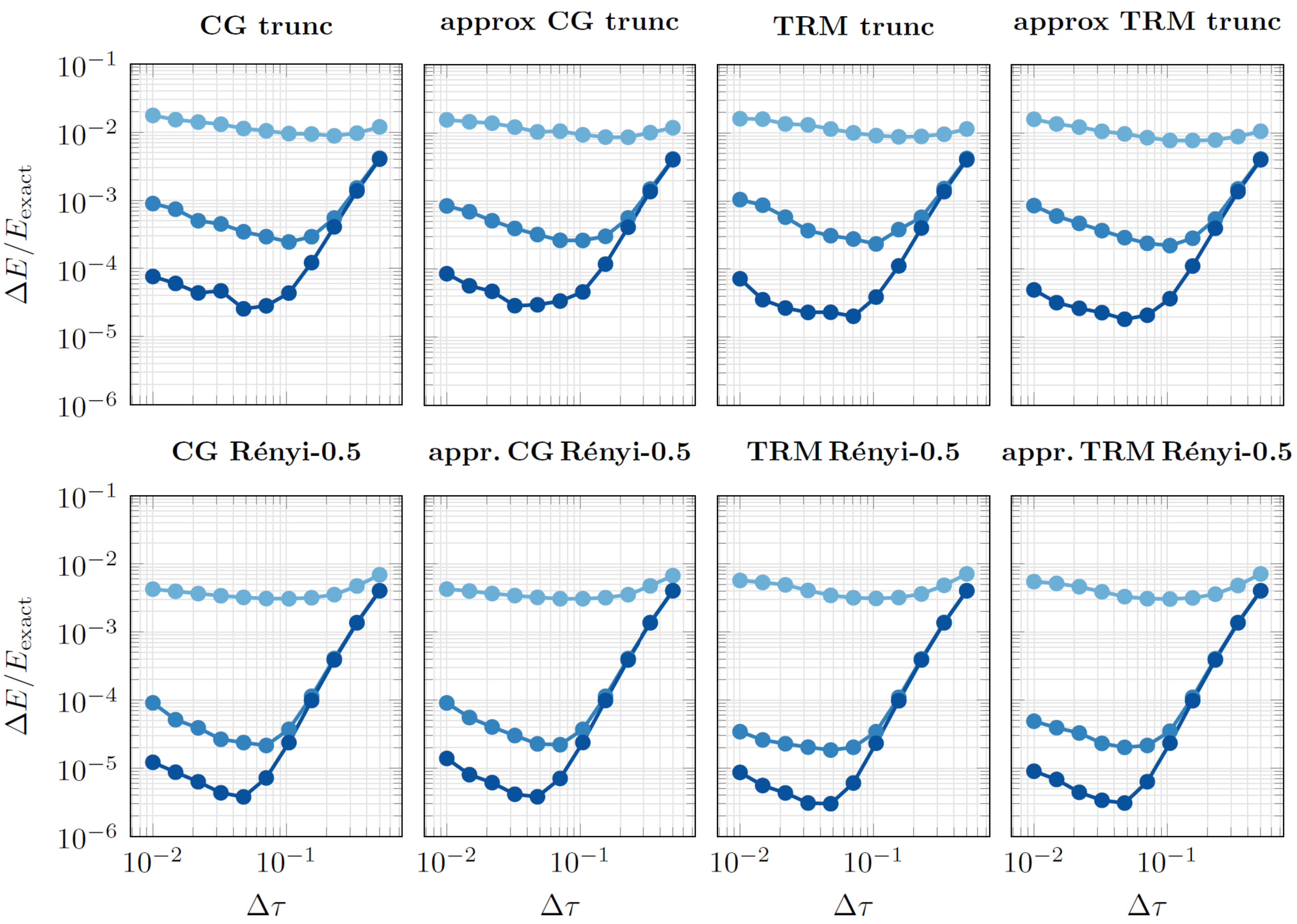
$$\langle\Psi|\hat{O}_{i,j}|\Psi\rangle =$$



TEBD2

$$\hat{U}(\Delta t) = \hat{U}^{\mathrm{TEBD2}}(\Delta t) + \mathcal{O}(\Delta t^3)$$




CG trunc
approx CG trunc
TRM trunc
approx TRM trunc
CG Rényi-0.5
appr. CG Rényi-0.5
TRM Rényi-0.5
appr. TRM Rényi-0.5
$\Delta E/E_{\mathrm{exact}}$
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
$\Delta\tau$
10^{-2}
10^{-1}

Ground state search logarithmic plot